MILITARY SPECIFICATION

SWITCHBOARD, TELEPHONE: SB-3614/TT, AUTOMATIC

This specification is approved for use by the U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a family of telephone switchboards which shall provide semi- and fully-automatic switching of local and distant calls on 2-wire and 2- and 6-wire lines and 2-, 4-, and 6-wire trunks utilizing space division switching matrices. Facilities shall be provided for interfacing with automatic central offices over 4-wire tone signaling trunks, 2-wire direct current (DC) closure trunks, and 6-wire recEive and transMit (E&M) trunks. The switchboard shall be configured so that it is convertible by means of modules plugged into a universally equipped housing to handle a variety of 2-, 4-, and 6-wire lines and trunks. The universally equipped housing shall be designated as switchboard housing SB-3614/TT. Modularly pluggable printed circuit boards (PCB) shall be as follows:

(a) 20-hertz (Hz) Ringdown Line/Trunk and Common Battery Supervision (CBS) Line (Type I PCB).

(b) Common Battery (CB) Dial Pulse, Dual-Tone Multi-frequency (DTMF) Line (Type II PCB).

(c) 4-wire Tone-Signaling Trunk (Type III PCB).

(d) 6-wire E&M Signaling Line/Trunk (Type IV PCB)

(e) DC Closure Trunk (Type V PCB)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters, U.S. Marine Corps (Code LMA-1, Washington, D. C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-T-704	- Treatment and Painting of Material
MIL-M-13231	- Marking of Electronic Items
MIL-F-14072	- Finish for Ground Signal Equipment
MIL-P-15024	- Plate, Tags and Bands For Identification Of Equipment
MIL-E-16400	- Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification for
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of
MIL-E-21981	- Electronic Type Designations, Identification Plates and Markings, Requirements for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-252	- Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	- Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-471	- Maintainability Demonstration
MIL-STD-781	- Reliability Tests Exponential Distribution
MIL-STD-785	- Reliability Program For Systems and Equipment Development and Production
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines
MIL-STD-965	- Parts Control Program

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extnet specified herein.

DRAWINGS

U. S. Marine Corps

775090A0349-1 - Switchboard, Telephone, Automatic

NATIONAL TECHNICAL INFORMATION SERVICE

AD 619666 Interference Reduction Guide for Design Engineers, Vol. I
AD 619667 Interference Reduction Guide for Design Engineers, Vol. II

(Application for copies should be addressed to the National Technical Information Service, P.O. Box 1553, Springfield, Va. 22151.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Prior to submission of any complete production units for acceptance, the contractor shall furnish the quantity of first articles specified (see 6.2) for testing under the direction and supervision of the procuring activity inspector to determine conformance to this specification. First article is defined as a switchboard which will be identical to production units which the contractor will subsequently produce in fulfillment of the contract. Approval of the first article by the activity concerned shall not relieve the contractor of the responsibility to supply a switchboard conforming to this specification. Examination and testing shall be as specified herein. Any changes or deviations of the production units from the first article shall be subject to the approval of the contracting officer. The first article may be included in the final shipment as part of the total quantity to be supplied under the contract, provided it has been completely refurbished.

3.2 Materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In all cases, general material requirements as to flammability, fungus-resistance, and nontoxicity as required by MIL-E-16400 shall be met.

3.2.1 Standard parts. Parts and materials approved in MIL-E-16400 shall be considered as standard.

3.2.2 Nonstandard parts. Electronic components, parts, devices, materials, techniques and technologies shall be in accordance with

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MIL-E-16400. Parts and materials not approved in MIL-E-16400 will be considered as nonstandard. Nonstandard parts approval must be received from the contracting officer. Compliance to MIL-E-16400 and MIL-STD-965 in nonstandard part usage is required.

3.3 Design. The switchboard shall conform to Drawing No. 775090A0349-1 and the requirements specified herein. The switchboard system shall consist of one or more switchboards. The switchboard shall be capable of individual operation as a 30-terminal switchboard; two switchboards may be combined to operate as a 60-terminal switchboard, or three switchboards may be combined to operate as a 90-terminal switchboard. The switchboard shall be a modular 30-line/trunk, lightweight, ruggedized, reliable, cordless, semi- and fully-automatic electronic equipment. It shall provide for the following:

(a) A centralized keysender control facility.

(b) Automatic features such as precedence, operator queuing for up to 15 terminations, preemption, reentry, call intercept, trunk directory, and trunk hunting as described herein.

(c) An operation interface with any of the following signaling modes.

(1) 4-wire tone signaling

(2) 2-wire DC loop signaling

(3) 6-wire E&M signaling

The capability shall be provided to allow the operator to establish up to two 10-party conference connections. Any conferee shall be able to release himself without affecting the existing connection. There shall be two levels of precedence, routine and priority, for access to the operator's queue, for call establishment and for preemption of connections. It shall be possible to designate individual subscribers as privileged. This action shall be accomplished by the operator either once, by predesignating such, or by assignment at the moment of each call extension. A CB DTMF subscriber having been predesignated as a priority subscriber may automatically complete a priority call by digiting R or P prior to the called party digits. Priority extended calls may preempt routine extended calls. The operator shall be provided means of predesignating call extensions to any line termination as intercept. Extension of calls to designated terminations shall be automatically intercepted into operator's queue for servicing. The switchboard shall be compatible with the equipments shown in Table I. The particular type of PCB applicable to the termination interface is indicated in Table I.

3.3.1 Functional requirements. Functional requirements of the switchboard shall consist of the following.

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TABLE I. Switchboard compatibility with military equipment.

Equipment	2-wire ringdown line	Type I PCB 2-wire CBS line	2-wire ringdown trunk	Type II PCB 2-wire CB line	Type III PCB 4-wire tone signaling trunk	Type IV E&M Line	Type IV E&M Trunk	Type V DC Closure
Telephone set TA-312/PT	X	X						
Telephone set TA-236/PT				X				
Telephone set TA-838/PT, 2-wire CB DTMF				X				
Switchboard telephone manual SB-22/PT			X					
Switchboard telephone manual SB-86/PT			X					
Switchboard telephone cordless manual SB-3082()/PT			X		X			
Switchboard telephone automatic SB-3614()/TT			X		X		X	
AN/TTC-38, AN/TTC-39 central office telephone automatic			X		X			
Signal converter CV-425	X		X					
Converter, telephone signal CV-1918, CV-1919, CV-2875					X			
Terminal sets, terminal telegraph AN/PCC-1, AN/VCC-1, AN/VCC-2	X		X		X			
Terminal telegraph TH-85/GCC	X							
Signal converter CV-1548			X					
Switchboard telephone manual SB-22PT/TA-977			X					
Switchboard telephone automatic SB-3614()/TT							X	
E&M OFF-Premise Subscriber terminal						X		
Dial Central Office (PTT)								X

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3.3.1.1 Terminal subsystem. The terminal subsystem shall be made up of the signal entry facility and line or trunk terminations. Each switchboard shall be configured to contain up to 30 lines or trunk terminations provided on a two-termination-per-printed-circuit-board basis (15 termination PCB's). Each of the 15 switchboard termination PCB slot positions shall accept the 2-wire ringdown line/trunk or CBS line PCB (defined as a type I PCB), or the CB dial pulse or DTMF line PCB (defined as a type II PCB). Up to 9 such positions shall accept the 4-wire tone-signaling trunk termination PCB (defined as a type III PCB), or 6-wire E&M line/trunk termination PCB (defined as a type IV PCB), or 2-wire DC closure trunk termination PCB (defined as a type V PCB). The signal entry facility shall consist of quick-disconnect connectors provided on the rear panel of the switchboard. The switchboard connectors shall provide cable terminations for up to 30 lines or trunks, of which 18 may be 2-, 4-, or 6-wire and the remaining may be 2-wire types. The connector shall also provide access to a rear-panel-mounted termination box. This termination box shall provide hole-in-head terminal mounting posts and high-voltage protection devices. Protection from extraneous high voltages for personnel engaged in operation and maintenance shall be provided by ground lugs on the switchboard.

3.3.1.2 Tone generation and detection subsystem. The tone generation subsystem shall provide for all machine- and operator-initiated signals necessary to supervise, signal, and service all types of terminations provided for by the switchboard system. The tone-detection subsystem shall provide automatic detection capability consistent with proper supervision and signaling of all line and trunk equipments for which termination will be provided within the switchboard. The frequencies, levels, and timing particulars of all signals generated and detected by the tone generation and detection subsystem shall be as specified in Tables II through VII.

3.3.1.3 Operator subsystem. The operator subsystem shall consist of the headset, all controls, and visual and audible indicators necessary to provide the basic operator functions listed below to line and trunk subscribers of the switchboard:

(a) Call Answer

(b) Call Extension including Conferencing and Test Tone

(c) Rering

(d) Unanswered Party Release; Answered Party Release (Manual Preemption)

(e) Call Monitoring

(f) Call Initiation including Conferencing and Test Tone

(g) Terminal Status Indicators

(h) One Time Priority Mark

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TABLE II. Switchboard originated signals (4-wire).

Function	Frequency (Hz)	Frequency tolerance	Send signal level	Duration	Remarks
Digit 1 2 3 4 5 6 7 8 9 0	697/1209 697/1336 697/1477 770/1209 770/1336 770/1477 852/1209 852/1336 852/1477 941/1336	±1.5%	Decibel referred to 1 milliwatt (dBmW) -7 ±2 decibel (dB) single tone level	Operator DTMF subscriber controlled	Operator/DTMF subscriber originated
Recall and line priority	941/1209				
Conference	941/1477				
Restart	852/1209	±1.5%	-7 dBmW ±2 dB	Continuous until interdigit is detected	Remains on until released by AN/TTC-38 if AN/TTC-38 forwards a disallowed tone burst sequence
Seize to tone signaling trunk	2250	±1.5%	-7 dBmW ±2 dB	Continuous maximum (max) of 2.5 ±0.5 second (s) or 475 +50 milliseconds (ms) after seize acknowledge (sz ack)	
Wait-in-queue	500	±2.5%	-14 dBmW ±2 dB	Continuous 2 s on) 2 s off) nominal 2 s on) until operator answers	Automatic
		±2.5%	-14 dBmW ±2 dB	Single sequence:	Automatic

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TABLE II. Switchboard originated signals (4-wire). (Cont'd)

Function	Frequency (Hz)	Frequency tolerance	Send signal level	Duration	Remarks
Preempt	1000	±2.5%	-14 dBmW ±2 dB	Single burst 0.25 s on) 0.25 s off) nominal 0.25 s on)	Automatic
Dial tone	500	±2.5%	-14 dBmW ±2 dB	Continuous (7.5 s max)	Terminated by first digit
Sz ack, release acknowledge (rl ack)	570	±1.5%	-7 dBmW ±2 dB	Until seize/release stops +250 ms minimum (min)	Inhibited after maximum below 2.5 ±0.5 s for sz ack 1.5 + 0.5 s for rl ack
Release to tone signaling trunk	2600	±1.5%	-7 dBmW ±2 dB	Continuous	Max of 3.5 ±0.5 s or until tripped
Test tone	1050	±2.5%	-4 dBmW ±0.5 dB	Continuous	Automatic
Priority(P) 1/ Immediate (I) Flash(F) Flash Override (FO)	941/1633 852/1633 770/1633 697/1633	±1.5%	-7 dBmW ±2 dB Single tone level	Operator controlled	Operator originated
Busy	500	±10%	-14 dBmW ±2 dB	Continuous interrupted sequence 0.5 s on) 0.5 s off) nominal	Automatic
Error	1000	±10%	-14 dBmW ±2 dB	Continuous interrupted sequence 0.25 s on) 0.25 s off) nominal	Automatic
Answer tone	2600	±1.5%	-7 dBmW ±2 dB	Single burst 275 ms ±50 ms	Automatic

1/ Priority calls into AN/TTC-38 by:

R1 = Priority R2 = Immediate R3 = Flash R4 = Flash override

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TABLE III. Switchboard received signals (4-wire).

Function	Frequency (Hz)	Frequency tolerance	Received signal level	Duration	Remarks
	The switchboard system shall receive the following 570-Hz, supervision and control signals				
sz ack rl ack	570	±1.3%	0 dBmW to -20 dBmW	Continuous	Until seize, release plus 250 ms or up to 3-s timeout
Seize	2250	±1.3%	0 dBmW to -20 dBmW	Continuous	1. Unitl sz ack or 2 s for converter trunk timeout 2. Until restart or 10 s for 3-digit tone-burst trunk timeout
Release	2600	±1.3%	0 dBmW to -20 dBmW	Continuous	Until rl ack or 2 s timeout on converter tone trunks and 3 s timeout on 3-digit tone burst trunks
Digit 1 2 3 4 5 6 7 8 9 0 (Priority)	697/1209 697/1336 697/1477 770/1209 770/1336 770/1477 852/1209 852/1336 852/1477 941/1336 941/1209	±1.5%	0 dBmW to -20 dBmW	Automatic office/ DTMF subscriber controlled	Automatic Office/DTMF subscriber originated
Interdigit	941/1633	±1.3%	0 dBmW to -20 dBmW	Continuous	Until end of restart or 3 s timeout

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TABLE III. Switchboard received signals (4-wire). (Cont'd)

Function	Frequency (Hz)	Frequency tolerance	Received signal level	Duration	Remarks
	The operation of the switchboard system shall be compatible with the following service signals which may be present during a connection to an automatic switchboard.				
Ringback (Normal)	425	±2.0%	-10 dBmW max	Same as normal ring	Same as normal ring
Ringback (Priority)	570/425	±2.0%	-10 dBmW max	Same as priority ring	Same as priority ring
Dial tone	425	±1.3%	-10 dBmW max	Continuous (10 s max)	Shut off by first digit
Busy	425	±1.3%	-10 dBmW max	1/4 s on 1/4 s off (continuous sequence)	10 s timeout or shut off by release
Preempt	570/425	±1.3%	-10 dBmW max	2 s ±1/4 s	
Error	425/1050 Frequency shift	±1.3%	-10 dBmW max	Continuous	1/8 s 425 1/8 s 1050
In-queue	425	±1.3%	-10 dBmW max	3 min or until answered	2 s on 4 s off 50 ms on 50 ms off
Answer	2600	±1.3%	-5 dBmW max	275 ±50 ms burst before connection	Indicates, to subscriber, connection to an automatic switchboard
Test	1050	±2.5%	-3.5 dBmW max	Continuous	Manual patch initiation at automatic switchboard
	The following signals may be present during a connection to an automatic switchboard. They are not required for control functions or service indications at the switchboard system.				
Conference	941/1477	±1.3%	-5 dBmW max	Subscriber controlled	

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TABLE IV. Switchboard received signals (2-wire).

The following signals may be received from a 2-wire DTMF subscriber and intended as signaling control.

Function	Frequency (Hz)	Frequency tolerance	Received signal level	Duration	Remarks
Digit 1	697/1209	±1.3%	0 to -20 dBmW per single tone	DTMF subscriber controlled	DTMF subscriber initiated
2	697/1336				
3	697/1477				
4	770/1209				
5	770/1336				
6	770/1477				
7	852/1209				
8	852/1336				
9	852/1477				
0	941/1336				
(Priority)	941/1209	±1.3%	0 to -20 dBmW		

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TABLE V. SB-3614 (XN-1)/TT 2-wire signals.

Function	Frequency (Hz)	Frequency tolerance	Send signal level	Receive signal level	Send sequence	Remarks
Ring to CBS ringdown line	20	±10%	Root mean square (rms) 90 volts (V) ±10%		2 s on 2 s off nominal 2 s on	Automatic
Ring to CB dial DTMF	20	±10%	rms 90 V ±10%		2 s on 2 s off nominal for 16 s	Automatic
Seize and release from ringdown line	20			12 V rms 16-25 Hz min	Continuous burst	Subscriber-controlled
Trunk signal	20	±10%	90 V rms ±10%	12 V rms 16-25 Hz min	2 s ±10% burst	Automatic
Ringback	500	±2.5%	-14 dBmW +6 dB -3 dB		Synchronous with line or trunk ring	Automatic
Dial tone	500	±2.5%	-14 dBmW +6 dB -3 dB		Continuous until 1st digit or 7.5 ±1/2 s for DTMF subscribers	Automatic
Wait-in-queue	500	±2.5%	-14 dBmW +6 dB -3 dB		2 s on 2 s off nominal 2 s on Until operator answers	Automatic
Preempt	1000	±2.5%	-14 dBmW ±2 dB		Single sequence 0.25 s on 0.25 s off 0.25 s on (Nominal)	Automatic

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TABLE V. SB-3614 (XN-1)/TT 2-wire signals. (Cont'd)

Function	Frequency (Hz)	Frequency tolerance	Send signal level	Receive signal level	Send sequence	Remarks
Test tone	1050	±2.5%	0 dBmW ±0.5 dB		Continuous	Automatic
Busy tone	500	±10%	-14 dBmW +2 dB		Continuous 1/2 s on 1/2 s off	Automatic
Off-hook warning to CBS, DP or DTMF CB lines	3125	±2.5%	Greater than 3.5 dBmW		Continuous interrupted sequence 0.5 s on 0.5 s off (Nominal)	Automatic
Error	1000	±10%	-14 dBmW ±2 dB		Continuous interrupted sequence 0.25 s on 0.25 s off (Nominal)	Automatic

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TABLE VI. Switchboard received signals (6-wire).

Function	Signal (Hz)	Frequency tolerance	Received signal level	Duration	Remarks
Dial tone	425	± 1.3%	-10 dBm max.	Continuous (10 s max.)	Shutoff by first digit
Busy	425	± 1.3%	-10 dBm max.	Continuous sequence 0.25 s on 0.25 s off	10-second timeout or shutoff by release
Preempt	475/425	± 1.3%	-10 dBm max	2.00 s ± 0.25 s	
Error	425/1050 frequency shift	± 1.3%	-10 dBm max.	Continuous	0.125 s 425 Hz 0.125 s 1050 Hz
Wait-in-queue	425	± 1.3%	-10 dBm max.	3 minutes or until answered	2 s on 4 s off 50 ms on 50 ms off
Answer	2600	± 1.3%	-5 dBm max.	275 ± 50 ms burst before connection	Indicates to subscriber, connection to an automatic switchboard
Test	1050	± 2.5%	-3.5 dBm max.	Continuous	Manual patch initiation to automatic switchboard
The following signals may be present during a connection to an automatic switchboard. They are not required for control functions or service indications at the switchboard system.					
Conference	941/1477	± 1.3%	-5 dBm max.	Subscriber controlled	
Insecure warning tone	1000	± 2.5%	-10 dBm max. 100 ms on } nom 6 s off } nom	Continuous interrupted sequence	Indicates a connection to a non-secure trunk

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TABLE VII. Switchboard originated signals (6-wire).

Function	Signal (Hz)	Frequency tolerance	Send signal level	Duration	Remarks
Digit 1	697/1209	± 1.5%	-7 dBm ± 2 dB single tone level	Operator/DTMF subscriber controlled	Operator/DTMF subscriber originated
2	697/1336				
3	697/1477				
4	770/1209				
5	770/1336				
6	770/1447				
7	852/1209				
8	852/1336				
9	852/1477				
0	941/1336				
Recall and line priority(R)	941/1209				
Conference (C)	941/1447	± 1.5%	-7 dBm ± 2 dB single tone level	Operator/DTMF subscriber controlled	Operator/DTMF subscriber originated
Seize	Ground (-48V) on M-lead	N/A	N/A	Duration of call	Automatic
Seize ack	Ground (-48V) on M-lead	N/A	N/A	Duration of call	Automatic
Release	Open (gnd) on M-lead	N/A	N/A	Duration of call	Automatic
Release ack	Open (gnd) on M-lead	N/A	N/A	Duration of call	Automatic

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TABLE VII. Switchboard originated signals (6-wire). (Cont'd)

Function	Signal (Hz)	Frequency tolerance	Send signal level	Duration	Remarks
Immediate (I)	852/1633	± 1.5%	Single tone level	Operator controlled	Operator originated
Flash (F)	710/1633	± 1.5%	Single tone level	Operator controlled	Operator originated
Flash Override (FO)	697/1633	± 1.5%	Single tone level	Operator controlled	Operator originated
Busy	500	± 10%	-14 dBm ± 2 dB	Continuous interrupted sequence 0.5 s on 0.5 s off } nominal	Automatic
Error	100	± 10%	-14 dBm ± 2 dB	Continuous interrupted sequence 0.25 s on 0.25 s off } nominal	Automatic
Dial pulse digiting	Nominal make-break of 60/40 ms	N/A	N/A	Ground and open connections	Automatic on M-lead
Insecure warning tone	1000	± 2.5%	-14dBm ± 2db	Continuous interrupted sequence 100 ms on 6 s off } nominal	Starts within 250 ms after connection to a insecure trunk is made

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TABLE VII. Switchboard originated signals (6-wire). (Cont'd)

Function	Signal (Hz)	Frequency tolerance	Send signal level	Duration	Remarks
Wait-in-queue	500	± 2.5%	-14 dBm ± 2 dB	Continuous interrupted sequence 2 s on, 2 s off } nominal until operator answers	Automatic
Ringback	500	± 2.5%	-14 dBm ± 2 dB	Synchronous with line or trunk ring	Automatic
Preempt	1000	± 2.5%	-14 dBm ± 2 dB	Single burst 0.25 s on, 0.25 s off } nominal	Automatic
Test tone	1050	± 2.5%	0 dBm ± 0.5 dB	Continuous	Operator originated
Priority(P)[1]	941/1633	± 1.5%	-7 dBm ± 2 dB	Operator controlled	Operator originated

[1]Priority calls into AN/TTC-38 by:

R1 = Priority
R2 = Immediate
R3 = Flash
R4 = Flash Override

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(i) Reentry Control

(j) Call Holding

(k) Privileged Assignment Control as follows:

(1) Priority Call Placement for DTMF terminations and preemption

(2) Intercept for any termination

(3) Trunk/Group Directory

(4) Special Trunk Supervision and Monitoring

(l) Terminal Hunting Controls

The operator subsystem also shall provide for the display of major status. Indicators shall be provided which show the operating status of the following:

(a) The control subsystem

(b) The power subsystem

(c) The 20-Hz generator

3.3.1.4 Matrix subsystem. The matrix subsystem shall be made up of a switching matrix network providing for nonblocking interconnection of any 30 terminations simultaneously, over 15 independent network paths and, with addition of plug-in PCB's, interconnection of any 60 terminations simultaneously over 30 independent network paths. In a 90-termination switchboard configuration, the switch matrix shall provide for simultaneous interconnection of any 60 of the 90 terminations over 30 independent network paths. The matrix shall provide for interconnection of all terminations, interconnection to the operator, and connection of service and signal tone generators and detectors to any termination.

3.3.1.5 Control subsystem. The control subsystem shall perform automatically the common system time-shared functions of all processing and service requests monitoring, using a stored program. It shall comprise a single processor, a main timing unit, read-only, random-access, and nonvolatile, alterable memories. It shall control all displays presented to the operator, and react to each of his requests entered by means of his panel controls. It shall perform a self-check for catastrophic failure and shall provide a failure indication through the operator subsystem indicator display. It shall respond to a 3-digit numbering plan to identify and extend to terminations, to access operator queue and trunk directory tables. The criteria shown in Table VIII shall be used for processing.

3.3.1.6 Power subsystem. The power subsystem shall provide for operation of the switchboard from an external DC prime power source;

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TABLE VIII. Numbering plan criteria.

First digit	Remaining digits	
0	None	Route to operator's queue
R/P	NXX[1/]	Extend, if privileged, as priority call; otherwise, return error tone
1-6	XX	Extend to XX
7,8	XX	Route to operator queue
9	XX	Go to trunk directory table. If calling termination is CB dial pulse, route to operator queue except when XX is to a distant manual board.

1/ N = 1-6 Only

DC-to-DC conversion shall be used to derive the necessary operating voltages for all subsystems.

3.3.2 Electrical requirements. Electrical requirements for the switchboard shall be as follows:

3.3.2.1 Terminal subsystem electrical requirements. The terminal subsystem shall provide terminations for the following line and trunk circuits.

3.3.2.1.1 Line/trunk interfaces. The 30 line/trunk circuits of the switchboard shall be selectable to operate as 30 2-wire line/trunk circuits or any combination of 30 line/trunks wherein no more than 18 designated terminations will be 2-wire DC closure trunks, 4-wire tone-signaling trunks, or 6-wire E&M trunks, and the remainder will be 2-wire ringdown line/trunk or 2-wire CBS, CB dial or CB DTMF lines. When 6-wire line/trunks are used, two 4-wire designated terminals will be occupied per 6-wire line/trunk. Selection of required line or trunk termination shall be made by utilization of pluggable PCB's. Each line/trunk PCB slot in the switchboard shall be universally wired to accept any 2-wire line/trunk PCB's. The 4-wire tone-signaling, 6-wire E&M trunk and DC closure trunk PCB's shall be placed in up to nine designated line/trunk PCB slots of the switchboard. Termination groupings desired are indicated below by types of pluggable PCB's. The mix of termination PCB's required shall be provided as specified (see 6.2):

(a) 20-Hz ringdown line/trunk and CBS line PCB (type I)

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(b) Common battery dial pulse or DTMF line PCB (type II)

(c) 4-wire tone signaling trunk PCB (type III)

(d) 6-wire E&M line trunk PCB (type IV)

(e) 2-wire DC closure trunk PCB (type V)

Within a PCB type having more than one termination mode, the change from one mode of operation to another shall be accomplished by switches on the PCB provided in a one-to-one correspondence with each line/trunk type. Each of the three termination-type PCB's shall operate in one of several modes. The type I PCB shall contain two independent terminations, each of which shall operate as follows:

(a) A 2-wire 20-Hz ringdown line or

(b) A 2-wire 20-Hz ringdown trunk or

(c) A 2-wire CBS line

The type II PCB shall contain two independent terminations, each of which shall operate as follows:

(a) A 2-wire CB dial pulse line or

(b) A 2-wire CB DTMF line

The type III PCB shall contain two independent terminations, each of which shall operate as follows:

(a) A 4-wire tone signaling converter trunk or

(b) A 4-wire 3-digit tone-burst trunk

The type IV PCB shall contain two independent terminations one of which shall operate as follows:

(a) A 6-wire E&M dial pulse trunk

(b) A 6-wire E&M DTMF trunks

(c) A 6-wire E&M dial pulse line

(d) A 6-wire E&M DTMF line

The other of which shall operate as follows:

(a) A 2-wire CB dial pulse line

(b) A 2-wire CB DTMF line

The type V PCB shall contain two independent terminations each of which shall operate as follows:

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(a) A 2-wire DC Closure dial pulse trunk or

(b) A 2-wire DC Closure DTMF trunk

Changing from one mode to another on a PCB type shall be possible without disrupting operation of the switchboard. Functions performed by the line/trunk interfaces shall include the following:

(a) Detecting a request for service, seize signal or ring signal

(b) Providing requisite notification to the common control or other appropriate subsystem that a request for service has been received

(c) Providing the requisite electrical interface with the matrix subsystem

(d) Sending ringback signal to the calling parties as required during the ring sequence

(e) Sending ring signal to a called party in response to the appropriate control signal

(f) Detecting an incoming call complete, or release, signal; and regeneration and transmission of such signal to the other connected termination if it is a trunk

(g) Sending answer tone to the TTC-38 when an offhook answer condition is detected or call is answered by operator on 3-digit tone burst trunks only.

3.3.2.1.1.1 Type I PCB in the 2-wire ringdown mode. In the 2-wire ringdown mode, both the seize and release signals from line/trunks shall be bursts of 20 Hz. An incoming call from a 2-wire ringdown line/trunk shall be initiated by detection of a 20-Hz signal. Release of 2-wire ringdown connections shall also be initiated by detection of a 20-Hz signal from either line or trunk in the connection. The 20-Hz release signals shall not be transmitted back to the line or trunk on which a release signal was received, but shall be transmitted to the other terminal in the connection only if such terminal is operating in the trunk mode. After transmission of this release, the 2-wire trunk terminal circuitry shall not interpret as a request for service the receipt of a ring signal to this terminal within a predetermined time of 2-1/2 seconds (s). A distinctive tone of interrupted 500 Hz shall be given to a calling party during the wait-in-queue period. During the ring signal interval, a 500-Hz ringback tone shall be sent automatically to the calling party. Ring signal to a called line shall be 20 Hz in a single sequence, 2 s on, 2 s off, 2 s on. Ring signal to a called ringdown trunk shall be a single burst of 20 Hz for a nominal 2 s. Ringback signal to a terminal operating in any mode shall be 500 Hz, synchronized to the on/off sequence of the ring-forward signal. The switchboard shall respond to an incoming 20-Hz trunk seize in 3 different manners. The standard supervision sequence shall return wait-in-operator-queue tone to the incoming trunk seize. Incoming

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release shall be detected and the trunk returned to an idle state. The first special supervision sequence shall return dial tone to the incoming trunk seize. The switchboard shall assign a DTMF receiver to the trunk, collect, process incoming dual-tone signaling and extend calls in normal fashion. The switchboard originating the trunk call shall monitor the first digit (R or P) to check that the originating subscriber does not place a call above allotted precedence. If allotted precedence is exceeded, the originating switchboard shall release the trunk and return error tone to the originating subscriber. Release shall be as in the standard sequence. The second special supervision sequence shall treat incoming trunk calls in the same manner as the first special supervision sequence. On outgoing calls, the switchboard shall forward seize (20 Hz) to the called switchboard, and provide ring-back tone (RBT) to the calling party. The switchboard shall then consider the call complete and take no further action in the call processing sequence. Release shall be as in the standard sequence. The switchboard shall respond to an incoming 20-Hz line seize in two different manners. The standard supervision sequence shall return wait-in-operator-queue tone to the incoming line seize. Incoming release shall be detected and the line returned to the idle state. The special supervision sequence shall return dial tone to the incoming line seize. The switchboard shall assign a DTMF receiver to the line, collect, process, incoming dual-tone signaling and extend calls in the normal fashion. The switchboard will monitor the first digit dialed (R or P) to check that the subscriber does not place a call above his allotted precedence. If the allotted precedence is exceeded, the subscriber shall receive error tone. Release will be in the standard fashion. The switchboard, by means of an operator-controlled access to nonvolatile, alterable memory, shall be able to mark 20-Hz trunks as standard or either type of special signalling sequence and 20-Hz lines as standard or special.

3.3.2.1.1.2 Type I PCB in the 2-wire CBS mode. In the CBS mode, the seize and release (off-hook and on-hook) signals shall be respectively a DC closure and a DC open circuit at the user's telephone. The CBS circuit shall function with subscriber loops of 1,200-ohm loop resistance and 5,000-ohm leakage resistance when terminated in a TA-312 Telephone Set. Ring and ringback signals shall be the same as for a 20-Hz ringdown line.

3.3.2.1.1.3 Type II PCB with 2-wire CB dial pulse or DTMF line. In the CB type of operation, seize and release (off-hook, on-hook) signals are respectively a DC closure and a DC open circuit caused by the user's telephone. With a 24-volt direct current (VDC) prime voltage, the CB termination shall recognize 10 milliamperes (mA) or less as a DC open (on-hook) indication and 16 mA or more as a DC closure (off-hook) indication. Supervision and dial pulse terminations signaling shall be recognizable through DC closures and openings as generated by the subscriber. The voltage and current limiting circuitry of the CB termination shall be such that the telephones will derive a minimum of 26 mA for talking and signaling current over loops of up to 1-3/8 miles of WD-1 field wire (450 ohm loop including telephone instrument resistance). Dial pulse shall accept the calling subscriber's digiting, convert the signaling to

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binary information, and make it available to the processor. The processor, when scanning these CB terminals, shall collect all signaling digits, process them, and connect up the calling party to the called party, automatically sending to the calling and called party such supervisory tones as are necessary (i.e., ring, ringback, busy, error or wait-in-queue, off-hook warnings). No operator action shall be necessary for completion of CB, dial pulse or DTMF calls. Ring and ringback signals shall be 2 s on, 2 s off, for 16 s. After a connection has been established, any CB dial pulse or DTMF termination shall be capable of recalling its local operator by momentarily depressing and then releasing the telephone hook switch. If the hook switch is depressed for more than 1.5 s, it shall be considered a release indication and appropriate release action taken.

3.3.2.1.1.4 Type II PCB in the 2-wire CB DTMF line mode. Operation of the type II PCB in the 2-wire CB DTMF line mode shall be as specified in 3.3.2.1.1.3 except that detection of incoming signaling shall be accomplished by use of a separately provided tone-detection circuitry. This circuitry shall convert signaling information to binary form for further processing.

3.3.2.1.1.5 Type III PCB in the 4-wire tone-signaling trunk mode. Terminations shall be provided in the switchboard to enable operation with an automatic central office such as an AN/TTC-38, another switchboard or a DTMF converter over a 4-wire tone-signaling converter trunk. Terminations shall also be provided in the switchboard system to enable operation with an automatic central office such as the AN/TTC-38 and the AN/TTC-39 over a 4-wire, 3-digit tone-burst signaling trunk. Up to 18 converter or tone-burst signaling trunks shall be provided for, two per type III CB. Supervisory tones available to 2-wire calling subscribers shall be available to 4-wire subscribers, using these trunks as access to the switchboard. No operator actions shall be required to complete 4-wire trunk calls if the originating terminations have DTMF signaling capability.

3.3.2.1.1.5.1 Outgoing signaling - tone signaling. On an outgoing call from the switchboard, the switchboard shall forward seize tone, 2250 Hz, to the distant end. The distant end may be a DTMF converter, an AN/TTC-38, an AN/TTC-25, a switchboard 3082 or another switchboard 3614. The distant end should respond with sz ack, 570 Hz. The switchboard shall detect sz ack for 475 milliseconds (ms), stop seize, and complete the connection from the originating terminal to the called termination. If the outgoing call is to a DTMF converter, the call shall be connected to converter operator's position and ringback returned to the originating subscriber until the converter operator answers the service request. If the distant end is an AN/TTC-38 or another switchboard 3614, the distant end shall return dial tone and prepare to accept address digits from the originating subscriber. Th calling subscriber (or operator) shall be able to key in the precedence digit, if desired, and the desired address digits. Detection of the first digit by the distant end should stop dial tone, the remaining digits shall be collected and the call completed in the normal manner. The originating switchboard shall guard against unauthorized precedence by monitoring the calling subscriber's

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precedence digits (R or P for connection to a distant switchboard and R1 or P for a connection to a distant AN/TTC-38). If the calling subscriber requests a precedence when he has none, the originating switchboard shall release the trunk connection and return error-tone to the calling subscriber. Otherwise, the call shall be normally processed. The operator shall be provided with a capability of marking trunks connected to automatic boards for this precedence monitoring function. If the calling subscriber terminates the call by going on-hook, the switchboard shall send release tone, 2600 Hz, to the distant end. The distant end should respond to detection of release by returning r1 ack, 570 Hz. The switchboard shall detect r1 ack for a nominal 475 ms, stop release, and return the trunk to idle.

3.3.2.1.1.5.2 Incoming signaling - tone signaling. For an incoming call from a converter, a switchboard 3614 or an AN/TTC-38 to the switchboard, the signaling sequence is as follows: The distant end sends seize. The switchboard detects seize for 300 to 500 ms and responds with sz ack. The distant end shall detect sz ack and stop transmitting seize. The switch-board shall detect the end of seize, stop the sz ack after 475 ms, and then send dial tone. The distant subscriber or operator then may key in digits, causing the switchboard to complete the connection automatically. If the distant end forwards release to the switchboard, the switchboard shall detect release for 525 ms, split the link, continue to detect release, for a total of 900 to 950 ms and then respond with r1 ack. The distant end should detect the r1 ack, and should stop sending release. The switchboard shall detect the end of release, and stop sending r1 ack after a 250-ms minimum, terminating the release sequence. The trunk termination shall always acknowledge a release signal, regardless of the busy-idle status of the termination. Should release end when the link is split, the link shall be reconnected.

3.3.2.1.1.5.3 Outgoing signaling - 3-digit tone burst trunk. The switchboard may be connected to an AN/TTC-38 over a 3-digit tone burst trunk. Outgoing signaling from the switchboard shall be the same as for the converter trunk. The distant board shall return dial tone to the calling subscriber, allowing the subscriber to key in appropriate digits. Precedence monitoring by the originating switchboard shall apply as in the converter trunk.

3.3.2.1.1.5.4 Incoming signaling - 3-digit tone burst trunk. A call from the AN/TTC-38 to the switchboard shall be initiated by forwarding seize signal to the switchboard. Upon detection of seize for 300 to 500 ms, the switchboard shall return a restart signal. The restart signal shall cause the AN/TTC-38 to stop sending seize signal and to send a tone burst sequence consisting of the recall digit (R), if precedence is required, followed by a 1-digit address (O) or by a 3-digit address (XXX). Each digit duration shall be 70 ± 15 ms. Break duration shall equal digit duration. At the end of the tone-burst sequence, the AN/TTC-38 shall send the interdigit tone. Having recognized a valid address and upon detection of the interdigit tone, the switchboard shall stop sending restart signal. If a valid address has not been received, the switchboard shall route the call to the operator's queue. In the AN/TTC-38, if the end of restart is

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detected any time between the end of seize and before the beginning of interdigit tone, the trunk shall immediately be made idle and placed into release processing. If end of restart is detected after beginning of interdigit but before timeout, the AN/TTC-38 shall complete the connection from the originating terminal to the switchboard. When the switchboard has determined that the called party has gone off-hook or the operator answers the call from queue, answer tone shall be forwarded to the AN/TTC-38. In the event that the call from the AN/TTC-38 is processed through a tandem switchboard, the directly connected switchboard shall forward answer tone after the seize sequence has ended.

3.3.2.1.1.6 Type IV PCB 6-wire E&M signaling mode. Termination shall be provided in the switchboard to enable operation with an automatic central office, another switchboard, or an off-premise subscriber in either dial pulse or dual-tone signaling modes. Up to nine E&M trunks shall be provided for, one per type IV PCB. The remaining nine terminals shall be used as CB dial pulse or DTMF line terminations, signaling for these terminations shall be the same as described in 3.3.2.1.1.3 and 3.3.2.1.1.4. Supervisory tones available to 2-wire subscribers shall be available to 6-wire subscribers using these trunks as access to the switchboard. No operator actions shall be required to complete 6-wire DTMF calls if the originating terminations have DTMF signaling capabilities. No operator action shall be required to complete 6-wire dial pulse trunk calls if the originating terminations have DTMF or dial pulse signaling capabilities. Two types of E&M signaling shall be possible with the type IV PCB, contact E&M and true E&M. The supervision conventions for the contact E&M shall recognize open circuit of the E & M leads as an on-hook indication and shall recognize a ground on the E & M leads as an off-hook indication. The supervision convention for true E&M signaling shall recognize a grounded M lead and an open E pair as an on-hook indication and battery (externally supplied -48 VDC) on the M pair and a grounded E pair as an off-hook indication.

3.3.2.1.1.6.1 Incoming signaling. For an incoming call from a central office, another switchboard, or subscriber, the signaling sequence is as follows. The distant end shall send seize to the switchboard by providing an off-hook on the E lead for a nominal 120 ms. The switchboard shall respond to the seize by providing an off-hook on its M lead. The switchboard shall return wait-in-operator-queue tone for manual incoming lines/trunks or return dial tone and prepare to accept digits for automatic incoming lines/trunks. Incoming release shall be detected when the distant end provides an on-hook indication on the E lead for a period of more than 120 ms. The switchboard shall respond to the release by providing an on-hook indication on its M lead.

3.3.2.1.1.6.2 Outgoing signaling (trunk). An outgoing call from the switchboard to a central office or distant switchboard shall utilize the following signaling sequence. The switchboard shall seize a trunk by providing an off-hook indication on its M lead. The distant end shall respond to the seize by providing an off-hook indication on the E lead. The switchboard shall consider the trunk

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nonoperational if an off-hook indication is not detected within 3 s after sending a seize. The call shall be processed according to the capabilities of the distant switchboard. If dial tone is returned, the switchboard shall be able to signal with either dial pulse or DTMF digits. If DTMF signaling is used, the switchboard shall limit access to only terminals with the DTMF capabilities. The switchboard shall monitor the dialed (keyed) digits to guard against unauthorized precedence users. If an unauthorized precedence is used, the switchboard shall release the trunk connection and return error tone to the calling party. Outgoing release shall be accomplished by an on-hook indication on the M lead; the distant switchboard shall acknowledge the release by providing an on-hook indication on the E lead.

3.3.2.1.1.6.3 Outgoing signaling (line). An outgoing call from the switchboard to an off premise (op) subscriber shall utilize the following sequence. The switchboard shall notify the subscriber of a seize by sending four on/off hook sequences of a duration of 2 s on 2 s off on the switchboard's M pair. The op subscriber shall acknowledge the seize by providing an off-hook indication on the E pair. After completion of signaling, a voice connection is established. Release is accomplished by providing on-hook indications by either subscribers of the call.

3.3.2.1.1.7 Type V PCB 2-wire DC closure trunk mode. Terminations shall be provided in the switchboard to enable operation with commercial central office DC closure lines with either dial pulse or DTMF signaling capabilities. Up to 18 DC closure trunks shall be provided for, two per type V PCB. Supervisory tones available to subscribers shall be available to terminations using these trunks as access to the switchboard. The operator shall be required to complete DC closure trunk calls for those terminals without dialing capabilities.

3.3.2.1.1.7.1 Outgoing signaling. The switchboard will seize the central office by placing a DC closure across the terminal pairs. Upon detection of this closure, the central office should return dial tone and prepare to accept digits from the originating subscriber. Detection of the first digit should stop the dial tone, the remaining digits shall be collected and the call completed in the normal manner. Outgoing release from the switchboard shall be accomplished by the removal of the DC closure from the terminal pair.

3.3.2.1.1.7.2 Incoming signaling. The central office shall initiate a call to the switchboard by sending a 20-Hz signal of a voltage greater than 12 volts (V) root mean square (rms). Upon detection of this signal, the switchboard shall return wait-in-operator-queue tone to the incoming trunk. The switchboard shall not provide a DC loop closure to the central office until the operator answers the incoming call. The switchboard shall not detect an incoming release from the central office.

3.3.2.2 Tone generation/detection subsystem electrical requirements. The tone generation/detection subsystem shall generate and detect signals as follows:

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3.3.2.2.1 Tone generation. This subsystem shall consist of tone generators in the switchboard.

3.3.2.2.1.1 20-Hz ring generator. The ring generator shall provide a 90-V ± 10 percent peak-square-wave ring signal across a 5,000 ohm resistive load at 20 Hz ± 10 percent. When ringing any 2-wire 20-Hz ringdown load, the on cycle of the ring signal shall be a nominal 2 s in a 4-s period. The 20-Hz ring generator shall drive a maximum load equivalent to a 1,000-ohm resistor. The full load voltage shall be not less than 75 V peak-square-wave, and the no-load voltage shall not exceed 140 V peak-square-wave. When the 20-Hz generator is connected to a line/trunk, short circuit protection shall be provided. A 20-Hz generator failure indicator shall be provided.

3.3.2.2.1.2 Test-tone generator. A 1,050 ± 25-Hz tone generator shall be provided in the switchboard. The test tone shall be available at the test-tone generator test point and at a set of special binding posts at the rear of the switchboard. The test-tone at these points shall be 0 decibel at one milliwat (dBmW) ± 0.5 dB into a 600-ohm load measured at the 2-wire zero transmission-level reference point. The test-tone generator shall be assigned a number within the numbering plan, and the operator shall be able to connect the test tone, through the switchboard, to any one of the 30 line/trunk terminals. When the test-tone is connected through the switchboard to a line operating in the 4-wire tone signaling mode, the test-tone measured at the 4-wire transmission-level reference point shall be -4 dBmW ± 0.5 dB into a 600-ohm load. Removal of test-tone and release of the line/trunk circuit to which the test-tone generator had been connected shall be automatic on receipt of the appropriate release signal from the line/trunk circuit.

3.3.2.2.1.3 Preempt tone generator. The preempt tone level measured at the 2-wire transmission level reference point shall be 1,000 Hz ± 2.5 percent at a level of -14 dBmW ± 2 dB into a 600-ohm load. Preempt tone shall be provided when any 2-, 4-, or 6-wire termination is employed.

3.3.2.2.1.4 Error tone. In operating the switchboard, when line or trunk terminations are utilized, the operator shall receive an error tone when unassigned or illegal digits are enabled by use of the keysender. Unassigned terminal numbers shall be 3-digit number sequences not used to identify terminal numbers or the test-tone number. Illegal digits shall be those which are enabled which do not represent number groups which can be interpreted by the common control. The error tone shall be a continuous interrupted sequence of 1 kilohertz (kHz) ± 10 percent, 1/4 s on and 1/4 s off. The signal level of the error tone shall be -14 dBmW ± 2 dB into a 600-ohm load.

3.3.2.2.1.5 Wait-in-queue tone. The wait-in-queue tone shall be 500 Hz ± 2.5 percent at a level of -14 dBmW + 6 dB, -3 dB into a 600-ohm load in a continuous interrupted sequence of 2 s on, 2 s off. This tone shall continue until the service request is answered by the operator or terminated by a user releasing from queue.

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3.3.2.2.1.6 Ringback tone. The ringback shall be synchronized with the ring signal 500 Hz ± 2.5 percent at a level of -14 dBmW + 6 dB, -3 dB.

3.3.2.2.1.7 Dial tone. Dial tone shall be sent to all automatic dialing/keying calling subscribers when the seize sequence is complete. Dial tone shall be 500 Hz ± 2.5 percent continuous at -14 dBmW, +6 dB, -3 dB.

3.3.2.2.1.8 Seize tone. The switchboard system shall provide seize signals to distant automatic switchboards. The signal frequency shall be 2,250 Hz ± 1.5 percent source at a level of -7 dBmW, ± 2 dB into a load of 600 ohms measured at the 4-wire transmission-level reference point.

3.3.2.2.1.9 Acknowledge tone. The switchboard system shall provide a sz ack and a rl ack tone to distant automatic switchboards. The acknowledge frequency shall be 570 Hz ± 1.5 percent at a level of -7 ± 2 dBmW measured at the 4-wire transmission-level reference point into a 600-ohm load.

3.3.2.2.1.10 Release tone. The switchboard system shall provide a release tone signal to distant automatic switchboards. The release frequency shall be 2,600 Hz ± 1.5 percent at a level of -7 dBmW ± 2 dB measured at the 4-wire transmission-level reference point into a 600-ohm load.

3.3.2.2.1.11 Off-hook warning tone. The switchboard system shall provide an off-hook warning tone to CBS and CB line subscribers who inadvertently leave the handset of their telephone off hook. The off-hook warning tone shall be 3,125 Hz ± 2.5 percent at a level of greater than 3.5 dBmW into a load of 600 ohms. The tone shall be an interrupted sequence of 0.5 s on, 0.5 s off until the handset is placed on hook or until terminated by the operator.

3.3.2.2.1.12 Busy tone. The switchboard shall provide a busy tone to the calling party or to the operator and calling party when the operator or calling party keys the number of terminal which is busy connected, or, in the case of a trunk, waiting-in-queue. The busy tone shall be 500 Hz ± 2.5 percent at a level of -14 dBmW ± 2 dB into a load of 600 ohms in a continuous interrupted sequence of 0.5 s on, 0.5 s off until the calling party releases or the operator releases the called party. Whenever a trunk release is detected, the switchboard shall provide busy tone to the local subscriber. Busy tone shall be returned to any line which is idle busy.

3.3.2.2.1.13 Dual-tone multifrequency generator. Tone generators necessary to provide dual-tone signaling shall be included in the switchboard. Each frequency of the transmitted signal shall be within 1.5 percent of its nominal value. In each case, the signal level of each tone measured at the 4-wire transmission-level reference point shall be -7 dBmW ± 2 dB into a 600-ohm load. The tone frequencies are listed in Table II.

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3.3.2.2.1.14 Answer tone. The switchboard system shall provide answer tone signal to the AN/TTC-38 over 3-digit tone burst trunks. The frequency shall be 2600 Hz ± 1.5 percent at -7 dBmW ± 2 dB measured at the 4-wire transmission level reference point into a 600-ohm load. Duration shall be 275 ms ± 50 ms in a single burst.

3.3.2.2.1.15 In-secure warning tone. The switchboard shall provide an in-secure warning tone to terminals that are connected to designated in-secure trunks. The tone shall be heard by both parties of the connection. The operator shall have the capability to mark a trunk as secure or in-secure. The in-secure warning tone shall be 1000 Hz ± 2.5 percent at a level of -14 dBmW ± 2 dB into a load of 600 ohms in a continuous interrupted sequence of 100 ms on, 6 s off. The tone shall start a maximum of 250 ms after the trunk connection is established.

3.3.2.2.2 Signal and tone detection. The tone detection subsystem shall consist of tone detection within the switchboard.

3.3.2.2.2.1 Seize detection. Seize detection circuitry shall respond to a continuous tone of 2,250 Hz ± 1.5 percent at signal levels between 0 and -20 dBmW. This detection must not detect 2,400 Hz at any level. Switchboard system detection time shall satisfy the acknowledge response times specified in 3.3.2.1.1.5.2. Seize detection circuitry shall not respond to other received tones shown in Tables II and III.

3.3.2.2.2.2 Release detection. Release detection circuitry shall respond to a continuous tone of 2,600 Hz ± 1.5 percent at signal levels between 0 and -20 dBmW. This detection must not respond to 2,400 Hz at any level. Switchboard system detection time shall satisfy acknowledge response times specified in 3.3.2.1.1.5.2. Release detection circuitry shall not respond to other received tones shown in Table II including answer tone (2,600 Hz for 325 ms maximum).

3.3.2.2.2.3 Acknowledge detection. The 570-Hz tone detection circuitry shall respond to signal levels between 0 dBmW and -20 dBmW within the frequency 570 Hz ± 2 percent. The detection shall not respond to signal frequencies below -35 dBmW. Acknowledge detection time shall satisfy response times specified in 3.3.2.1.1.5.1.

3.3.2.2.2.4 Dual-tone multifrequency tone detection. Dual-tone multifrequency tone detection circuitry shall detect all DTMF digit and priority tones as shown in Tables III through VI under switchboard-received tones. Detection shall be provided for all tone levels from 0 dBmW to -20 dBmW per single tone and to DTMF frequencies within ± 1.5 percent of nominal. There shall be no detection of more than one dual tone listed in Tables III through VI at any one time.

3.3.2.2.2.5 20-Hz detection. Detection sensitivity of 20 Hz shall provide for reliable detection of 20-Hz signals of amplitudes of 12 V rms or greater, in the range of 16 to 25 Hz.

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3.3.2.3 Operator subsystem electrical requirements. The operator subsystem of the switchboard shall provide all facilities needed by the operator to service terminations. Facilities for answering, initiating, extending, preemption, reentry, monitoring and releasing calls shall be provided by use of standard controls. Facilities shall also be provided for entering, deleting or modifying trunk directory data, precedence, intercept and special trunk class mark information into nonvolatile, alterable memory. Entry of this information shall be by the operator's keyset.

3.3.2.3.1 Operator's headset. The operator's circuit shall be designed for compatibility with Headset H-182. Two terminals which parallel the operator headset circuit shall be provided at the rear panel. The operator shall be provided a volume control knob on the front panel to allow adjustment to the audio level provided in his headset.

3.3.2.3.2 Terminal status indicator. Visual indicators shall be provided, one for each terminal, to indicate to the operator the status of terminals. The terminal status indicator shall light in a steady fashion if the terminal is busy, flash 1 s on, 1 s off if the terminal is waiting in the operator's queue, flash 0.5 s on, 0.5 s off if the terminal has been determined out of service (Type III PCB only), flash 0.25 s on, 0.25 s off if the operator is connected to the terminal, and shall be off if the terminal is idle.

3.3.2.3.3 Call-night alarm. An audible call-night alarm, with linearly adjustable volume control, shall be provided. The alarm shall sound whenever there is a call waiting and the operator is idle. The alarm shall not be adjustable below a minimum audible signal.

3.3.2.3.4 Called-release control. A momentary action pushbutton type called-release control shall be provided. This control shall provide the capability to release any unanswered called party, including an individually called conferee, without releasing the calling party or any previously connected conferees. This control shall also act as a clear function when the operator's keysender is in a data change mode. The control shall light when it is used in data entry. Suitable marking shall indicate the dual function of this control.

3.3.2.3.5 Visual-indicator control switch. A visual-indicator control switch shall be provided. This control shall provide a brightness adjustment for all visual indicators and panel illumination lamps required to allow night operation. The switch shall enable lamp test and illumination on/off control.

3.3.2.3.6 Power control switch. A switch shall be provided to turn all power to the switchboard on or off. A lamp shall provide visual indication of the power on state.

3.3.2.3.7 Operator-release control. A momentary action pushbutton control shall be provided to enable the operator to release himself from any terminal or call-in-process.

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3.3.2.3.8 Conference select. A control shall be provided to allow the operator to select a conference circuit for any subscriber and subsequently called terminals. This control shall be the pushbutton C on the keyset. This button needs to be depressed only once for conference select. An indicator shall allow the operator to note when both conference links are busy.

3.3.2.3.9 Terminal-mode programming controls. Terminal-mode programming controls (toggles) shall be provided on a one-to-one basis with each of the 30 line/trunk terminals, to facilitate changing modes of the signal handling specified in 3.3.2.1.1.

3.3.2.3.10 Failure alarms. The operator shall be provided visual indications of a failure in the control, DC power input, or 20-Hz generator. Failure of the control subsystem shall also sound an audible alarm.

3.3.2.3.11 Call-answer control. Depressing the call-answer control (CALL ANS) momentary action button shall disconnect the operator from his present connection and either return him to idle (if no queuing exists) or connect him to the first terminal in queue. If the operator is idle, depression of the button shall effect a connection to the first queue position if someone is in queue. The queue shall hold a maximum of 15 terminals.

3.3.2.3.12 Call extension. Extension of an incoming call to any line/ trunk or establishing conference connection shall be performed by use of the operator keysender. The operator need not determine whether or not the called line/trunk is busy.

3.3.2.3.13 Priority mark. An extended call shall be marked priority by the operator, using the P or R switch in the keysender prior to entering the three digits which identify the called terminal.

3.3.2.3.14 Multiple ring (rering). The operator shall be able to multiple ring an unanswered terminal by rekeying the terminal digits. Each time the terminal is keyed, the appropriate line/trunk ring sequence shall be initiated, if a busy status is not detected.

3.3.2.3.15 Call status indicator. This lamp shall be steadily on if the operator's circuit is connected to any nonpriority line/trunk. It shall flash when the operator is answering priority-privileged terminals or monitoring a priority-marked call.

3.3.2.3.16 Operator keysender. An operator keysender shall be provided to enable the operator to perform functions necessary to service all types of subscriber terminations. The keysender shall perform two primary functions as follows:

3.3.2.3.16.1 Call processing function. The operator keysender shall consist of 16 momentary action buttons. There shall be one pushbutton for each of the digits 0 through 9; one for each of the precedence levels, priority, immediate, flash, and flash override; one conference pushbutton; and one recall pushbutton. The keysender shall

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act as a terminal director keysender for calls being initiated, monitored, or extended, and shall provide the means to address any assigned termination. The keysender shall automatically become a tone keyer when a call is extended to a terminal operating in the DTMF trunk signaling mode. When such a call is released from the operator's connection, the keysender shall automatically return to its terminal director status. A control pushbutton shall be provided to the operator to return the keysender to the terminal director status.

3.3.2.3.16.2 Data change function and control. The operator's keysender shall provide entry for a changeable data base into the switchboard system. Storage for this data base shall be provided by nonvolatile, alterable memory. The nonvolatile, alterable memory shall provide for a trunk directory translation table which shall relate terminal assignments to trunk groups. Storage shall be provided for a minimum of twelve trunk groups, each capable of containing up to twelve trunks. The memory shall also provide storage for precedence (priority only), and intercept classmarks on any one or all 30 (60 to 90 in a stacked switchboard configuration) switchboard terminations when the terminations are lines. When terminations are trunks, storage shall be provided for classmarking up to 18 independent trunks (36 or 54 in a stacked switchboard configuration) as special in that automatic supervision features shall be provided in place of normal manual supervision. The entry format for data shall utilize the precedence keys of the operator's keysender to enter, delete, erase or verify data. Terminal and trunk group designations shall be entered through use of digits 0 through 9. Verification of data entry or a check on already entered data shall be provided by audible and visual means. Error tone shall indicate invalid verification. Busy tone shall indicate successful verification.

3.3.2.3.17 Conference busy. A lamp shall indicate when no conference link is available.

3.3.2.3.18 Reentry/call holding. A switch to be designated "Reentry/ Call Holding" shall provide the operator the following capabilities:

3.3.2.3.18.1 Call holding. When the operator disconnects temporarily from a call on any incoming line or trunk prior to extending the call to the called line or trunk, by activating the reentry control, the termination shall be held. The operator shall be provided a visual/audible indication of a party on call holding identical to that for a priority subscriber in queue. The operator shall be able to initiate calls or monitor terminals while such a calling party is on hold. The operator shall be able to reconnect to the calling party on hold by operating the call answer control, and he shall then be able to extend the call in the normal fashion.

3.3.2.3.18.2 Reentry. The operator shall be able to mark a terminal for reentry after keying the called terminal number. A visual indication shall inform the operator that a terminal is marked for reentry. He may then initiate calls, monitor terminals, answer requests for service or extend calls while such terminal is marked

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reentry. The operator shall be able, subsequently, to reconnect to the marked terminal by again reactivating the reentry control.

3.3.2.3.19 Precedence pushbuttons. The priority (P), immediate (I), flash (F), and flash override (FO) pushbuttons shall be provided for use only when in the tone keyer and data change mode of operation. The priority (P or R) pushbutton shall provide the capability to designate a call as a priority call when the keysender is in the terminal directory mode.

3.3.2.3.20 Trunk and line hunting. Terminal hunting shall be provided for two groups of terminals consisting of terminals 23 through 26 and 27 through 30 on each switchboard in the system. Any terminal in any group may be programmed as unassigned, a 2-wire ringdown line, 2-wire CBS line, 2-wire ringdown trunk, or any 4-wire trunk type, 2-wire DC closure trunk, or any 6-wire E&M trunk or line type, as long as each terminal in any one group is programmed the same way. A switch shall be provided on the switchboard for each group to enable or inhibit hunting for that group. A call initiated to any terminal in an enabled trunk group shall be able to access any idle terminal of the group initiated to. When a trunk directory capability is added through use of nonvolatile, alterable memory, then at least 12 trunk groups shall be provided for, with up to 12 trunks per group. The trunk hunting algorithm used with the alterable memory shall search for a usable trunk from the highest to the lowest terminal number assigned in the trunk group. If a malfunctioning trunk is found in the group, the switchboard shall skip that trunk and proceed to the next available trunk in the group. Provision for group assignment shall be through operator access to nonvolatile, alterable memory.

3.3.2.3.21 Data change control. A guarded toggle switch shall be provided, to enable conversion of the operator's keyset into a nonvolatile, alterable memory data change unit. This toggle shall function only when the nonvolatile, alterable memory is inserted into the switchboard.

3.3.2.3.22 Data entry alarm. A visual and audible alarm (error tone to headset) shall be provided the operator upon incorrect entry of data. If the operator attempts to enter a trunk group that has already been entered or attempts to enter a trunk that already exists in another trunk group, busy tone shall be provided to the operator via headset.

3.3.2.3.23 Trunk routining. Trunk routining shall be provided for all four-wire tone signaling trunks. The routining shall be initiated under processor control once every 120 seconds for all non-busy, assigned type III PCB terminals. The routining sequence shall send the release signal (2600 Hz) and monitor the terminal for the presence of Release Acknowledge (570 Hz). If Release Acknowledge is not received within the allowed timeout (1.5 s), the switchboard shall mark the terminal idle, out-of-service. The switchboard shall signal the operator/maintainer that the terminal is out-of-service by flashing the appropriate terminal status lamp at a 1 Hz rate. The

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terminal once marked idle, out-of-service shall remain in that state until it either passes a subsequent routining attempt, successfully completes an incoming seize request, or successfully completes an outgoing call. The switchboard shall also mark a terminal as idle, out-of-service when a seize timeout occurs, either during an incoming or outgoing call attempt.

3.3.2.4 Matrix subsystem electrical requirements. The switchboard interconnecting matrix subsystem shall consist of the following elements:

3.3.2.4.1 Wideband and voice matrix. The switchboard shall contain a 2-wire space division matrix providing for nonblocking connection of up to 30 subscribers simultaneously, or, in its stackable arrangement, for nonblocking connection of up to 60 subscribers simultaneously. Two of the provided-for linking paths (14 and 15) between subscribers shall be specified as conference links, and shall also be used as nonconference links. One link, No. 13, shall be used to connect test tone to any appropriately connected subscriber. Link 13 shall also be used to connect any two subscribers. The 2-wire space division matrix shall provide for normal path connection between subscribers under control of the control subsystem, and provide a capability to pass wideband information. Through appropriate 2- to 4-wire conversion techniques, the matrix shall provide access to 4-wire trunks.

3.3.2.4.2 Signal generation and detection matrix. A single-ended space division matrix shall be provided to interconnect all subscribers to tone-generation bus systems and common-equipment detection circuitry. Tones for service and signaling information shall be provided on a bussing arrangement, and every subscriber shall be connected to every such bus.

3.3.2.5 Control subsystems electrical requirements. Common control circuitry shall integrate the several subsystems into one operating system. The common control shall interpret and implement all manually activated or automatic supervisory and control signals, and effect the automatic, or manually activated functions specified herein. A visual and audible indicator shall be provided to indicate a catastrophic failure in the common control subsystem. Catastrophic failure is defined as loss of system clock, a program memory access or proper instruction execution. The common control subsystem shall consist of the items specified in 3.3.2.5.1 through 3.3.2.5.3.2, and through use of these items shall be able to perform the following functions automatically:

(a) Terminal scanning for service

(b) Call queuing

(c) Call initiation

(d) Call extension

(e) Operator service requests

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(f) Call breakdown

(g) Precedence operation

(h) Class-marking operations

(i) Preemption operation

(j) Terminal group hunting

(k) Trunk directory data entry

(l) 4-wire trunk routining

3.3.2.5.1 Processor. A processor shall process the operational call program provided for in the stored program applicable to servicing terminations in 3.3.2.5. In the stacked configuration of the switchboard, the processor in any switchboard may be used as the operational master. Other processors shall be available to be utilized as spares to the master.

3.3.2.5.2 Memory. The operational program providing for all automatic processing shall be stored in solid-state read-only memory. Information stored in read-only memory shall be permanent and not lost due to power failure. Current status and housekeeping data needed to provide immediate switchboard processing shall be stored in solid-state random-access memory. Information shall not be required to be retained during power shutdown. Modifiable data base information such as trunk directory translation tables, trunk groups, precedence, and intercept classmarking shall be stored in nonvolatile, alterable memory. Access to this memory for purposes of update shall be through the operator's keysender. Information stored in this memory shall not be lost when power is shut down.

3.3.2.5.3 Program. The contractor shall provide call-flow information analysis diagrams which shall include types of signals interchanged and timing for all types of calls or services to any terminal or input/output device to ensure that a complete understanding has been reached between the procuring activity and the contractor. This documentation shall be reviewed at periodic intervals, at the request of either the contractor or the contracting officer's technical representative. Flow diagrams, table organization, and other descriptive literature defining the design structure and operating procedures of the computer programs of each subsystem in the common control software package shall be documented and provided to the procuring activity for review. All computer programs, debug and test routines, and microprograms developed for, or utilized by, the stored-program control processor and its associated components (switch-to-processor interface control, memory control, matrix control, etc.) shall be documented in accordance with the following: All programmer's source language coding, assembly language coding (if applicable), object language coding, and computer listings shall be provided to the procuring activity, together with keying information to the applicable flow charts for checking and

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understanding the operation and implementation of each program, microprogram or both. Complete functional specifications shall also be provided for each program, microprogram, or both to define in detail the structure, operation, and interoperation. Program entries and exits, data base accesses, timing of routines and subroutines, and processing requirements shall be included. Data base characteristics shall be included. Mnemonic or other appropriate labeling of all data items shall be provided, as necessary, to implement and utilize the required switch control processor data base in conjunction with the remainder of the software. The documentation shall be provided to the procuring activity as generated.

3.3.2.5.3.1 Software program design. Software program design documentation shall include a descriptive narrative of program objectives and interoperations, program entries and exits, data table accesses, timing of routines and subroutines, and processing requirements of each program. Narrative text shall be supplemented with general flowcharts of program and subprogram logic, and diagrams of data flow.

3.3.2.5.3.2 Data base design. Data base design is defined as the compilation of all permanent or semipermanent data items contained in, and accessed by, a computer program. Data base documentation shall identify the types and characteristics of all program data items, and shall describe the structural layout of the permanent or semipermanent data items contained in, and accessed by, a computer program. Mnemonic labeling of all data items shall be in accordance with the logical guidelines established in the program design documentation.

3.3.2.6 Power subsystem electrical requirements. The power subsystem shall provide for the operation of the switchboard from an external DC prime power source. Circuitry shall be included to provide for immediate changeover, without loss of any existing call connections, from external DC prime power to battery power if the DC prime voltage drops below the point where the switchboard functions normally. Operation from external prime DC voltages of 22 to 30 VDC shall be provided for. Power requirements shall be 24 VDC, at no greater than 4.7 amperes (A) for each switchboard. Whenever the power level of the battery in use falls below a predetermined level, an indication shall be provided to the operator.

3.3.2.6.1 Battery connection. Two sets of independent connections shall be provided to allow connection of batteries to the switchboard. Connection/disconnection of each battery shall be independent of the other and allow for continuous switchboard service during change.

3.3.2.6.2 Prime power connection. Connection of prime DC power to the switchboard shall be provided independently of battery connections.

3.3.2.6.3 Direct current power converter. The switchboard DC/DC power converter shall provide all necessary DC voltage to the operating subsystems of the switchboard, when provided with input DC voltages of 22 to 30 VDC. Protection shall be provided to prevent

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inadvertant reversal of DC polarity or voltages higher than 30 VDC from damaging the DC/DC converter.

3.3.3 Physical requirements. The general form factor shall permit efficient ground or vehicular shelter operation of the switchboard system. All controls, connectors, etc. must be accessible during operation of the expanded configurations.

3.3.3.1 Front panel. The front panel shall contain all frequently-used operator controls and indicators, arranged to focus operator attention on the essential controls and to provide maximum ease of operation. The operator's headset connector shall be located on the front panel of the switchboard.

3.3.3.2 Rear panel. All signal-line connection facilities shall be located on the rear panels or attachable to it. The cable, or cables, required for interconnecting the switchboard when establishing an expanded configuration shall be stored in the switchboard when not in use. All multipin quick connect connectors required for signal line connection or interconnection of switchboards for expansion shall be located on the rear panel.

3.3.3.3 Covers. Covers shall be provided for both the front and rear panels of the switchboard. These covers shall be attached to the switchboard for field transport or storage, and shall provide protection from impact, dust, sand, snow and rain.

3.3.3.4 Transit case. A transit case shall be provided which will house at least 16 spare PCB's. Provision shall be made to protect the stored PCB's from any physical or electrical damage when the transit case and its associated spare PCB's are subjected to the environmental tests specified in 4.6.10. The transit case shall be constructed so that it and its contents shall be functionally useful after being tested in accordance with 4.6.10. The transit case shall also contain sufficient space to store one PCB extractor and such running spares as lamps and fuses.

3.3.3.5 Printed circuit board extractor. A PCB extractor capable of extracting any switchboard PCB shall be provided.

3.3.4 Transmission requirements. Transmission requirements shall be as follows:

3.3.4.1 Two-wire. The switchboard shall meet the following 2-wire transmission requirements. Transmission characteristics measurements shall be made between any two of the thirty 2-wire terminal pairs or the audio pairs of any 6-wire terminal as measured on the rear panel of the switchboard. Unless otherwise specified herein, these measurements shall be made at -4 dBmW. Voice-band transmission characteristics shall correspond to frequencies between 300 Hz and 3.5 kHz. Wideband transmission characteristics shall correspond to frequencies between 3.5 kHz and 108 kHz.

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3.3.4.1.1 Voice band.

3.3.4.1.1.1 Terminal impedance. The impedance of any line/trunk looking into the switchboard shall be 600 ohms ± 10 percent, with a phase angle within the limits of 0° to + 10° in the frequency range 300 to 3,500 Hz.

3.3.4.1.1.2 Frequency response. On any connection through the switchboard, the frequency response relative to 1 kHz shall be ± 1.0 dB over the frequency band of 0.5 to 3.5 kHz, and shall be within the limit of +1 and -3 dB over the frequency band 0.3 to 0.5 kHz. The response above 3.5 kHz shall never exceed +1 dB relative to the 1-kHz response.

3.3.4.1.1.3 Longitudinal balance. The longitudinal balance of any line or trunk circuit shall be no less than 40 dB over the frequency range 0.3 to 3.5 kHz.

3.3.4.1.1.4 Harmonic distortion. The total harmonic distortion within the frequency band 0.3 to 3.5 kHz produced by transmission of any single frequency test signal through the switchboard shall be at least 30 dB below the fundamental of the transmitted frequency. Power line frequencies, harmonics of power line frequencies, and noise introduced through the power supply shall have an rms sum no less than 44 dB below the test signal level.

3.3.4.1.1.5 Insertion loss. Insertion loss shall be no greater than 3.5 dB at 1,000 Hz.

3.3.4.1.1.6 Noise level. The noise level of any voice transmission circuit through the switchboard shall not exceed 22 dB above reference noise with C message weighting (dBrnC) when the switchboard is terminated with 600 ohms.

3.3.4.1.1.7 Impulse noise. The noise level due to impulse sources on any voice transmission circuit through the switchboard shall be no greater than -40 dBmW for pulses enduring 1 microsecond or longer on any line/trunk due to switching on that line/trunk.

3.3.4.1.1.8 Voice frequency crosstalk. Measured terminal to terminal, the crosstalk coupling loss between voice frequency circuits within the switchboard shall be no less than 60 dB between equal signal-level points at any frequency within the range 0.3 to 3.5 kHz.

3.3.4.1.1.9 20-Hz crosstalk. With a 20-Hz 135-V peak signal being transmitted to a 2-wire terminated line/trunk, the line outputs of each nonoperating line/trunk shall be less than 45 dBrnC.

3.3.4.1.1.10 Envelope delay distortion. Measured terminal to terminal, the envelope delay distortion through the switchboard, excluding conference circuitry, shall not exceed 100 microseconds between any two frequencies within the range 0.6 kHz to 0.8 kHz. From 0.8 to 3.5 kHz, the delay shall not exceed 50 microseconds between any two frequencies.

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3.3.4.1.1.11 Absolute delay. The absolute delay of a voice signal in passing through the switchboard shall not exceed 300 microseconds.

3.3.4.1.1.12 Transmission level. The maximum recommendedd transmission level into and through the switchboard shall not exceed 0 dBmW. Levels in excess of +4 dBmW may be limited.

3.3.4.1.1.13 Output variation. With a tone applied at -4 dBmW, output variation on any line/trunk to any other line/trunk shall not exceed ± 0.5 dB.

3.3.4.1.1.14 Conference circuits. The insertion loss between any input and any other output shall be 3 dB ± 1 dB at 1,000 Hz. Total insertion loss with five line/trunks connected shall not exceed 13 dB at 1,000 Hz, and with ten line/trunks connected, the loss shall not exceed 20 dB at 1,000 Hz.

3.3.4.1.2 Wideband. Wideband transmission characteristics shall correspond to frequencies between 3.5 kHz and 108 kHz.

3.3.4.1.2.1 Frequency response (wideband). On any connection through the switchboard, 2-wire to 2-wire, the frequency response relative to 1 kHz shall be +1.0, -2.0 dB over the frequency band 3.5 kHz to 108 kHz.

3.3.4.1.2.2 Crosstalk. Measured 2-wire to 2-wire terminal, the crosstalk coupling loss between circuits within the switchboard shall not be less than 55 dB between equal signal-level points at any frequency within the range of 3.5 to 1.8 kHz. For this measurement, line termination shall be 150 ohms.

3.3.4.1.2.3 Envelope delay distortion (wideband). Measured 2-wire terminal to 2-wire terminal, the envelope delay distortion through the switchboard, excluding conference circuitry, shall not exceed 10 microseconds between any two frequencies within the range of 3.5 kHz to 108 kHz.

3.3.4.2 4-wire. Transmission characteristics for the 4-wire tone signaling terminals shall be as follows:

3.3.4.2.1 Insertion loss (4-wire). The insertion loss, 4-wire to 2-wire or 2-wire to 4-wire, shall not exceed 6 dB at 1,000 Hz.

3.3.4.2.2 Frequency response (4-wire). On any connection through the switchboard system, 4-wire to 2-wire or 2-wire to 4-wire, the frequency response relative to 1 kHz shall be + 1 dB -3 dB over the frequency band of 0.3 to 0.5 kHz. The response from 0.5 kHz to 50 kHz shall be ± 1 dB relative to the 1 kHz response.

3.3.4.2.3 Return loss. With reference to a 600-ohm resistive load, the return loss shall meet the following requirement without regard to whether the termination is idle or busy.

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(a) From 0.3 kHz to 30 kHz, the minimum return loss shall be 20 dB.

(b) From 30 kHz to 50 kHz, the minimum return loss shall be 15 dB.

3.3.4.2.4 Transhybrid loss. The transhybrid loss for any 4-wire terminal connected through the matrix to a 2-wire terminal which is terminated with 600 ohms shall be greater than 25 dB over the frequency range 300 to 3,500 Hz, and 20 dB over the frequency range of 3.5 kHz to 50 kHz.

3.3.4.2.5 Rise time. On any connection through the switchboard system, 4-wire to 2-wire or 2-wire to 4-wire, the signal rise time shall be no less than 0.1 V per microsecond when terminated in 600 ohms.

3.3.4.2.6 Crosstalk. Measured 4-wire to 4-wire terminal, the crosstalk coupling loss between circuits within the switchboard shall not be less than 55 dB between equal signal level points at any frequency within the range of 3.5 kHz to 50 kHz. For this measurement, the line termination shall be 150 ohms.

3.3.4.2.7 Maximum signal. On any connection through this switchboard system, 4-wire to 2-wire or 2-wire to 4-wire, there shall be no hard limiting of signal amplitude for input signals with less than 5.0 V, peak to peak.

3.3.5 Safety requirements. Electronic system safety provisions shall be in accordance with requirements 1 and 8 of MIL-STD-454. The contractor shall conduct safety studies, perform preliminary hazard analysis, and prepare a safety statement for the switchboard.

3.4 Reliability. Reliability of operation shall be considered of prime importance in the design and manufacture of the switchboard. The contractor shall employ all methods possible in the process of design, development and manufacture which will assure quality and maximum reliability consistent with the state-of-the-art. In the functional application of parts, adequate factors of safety shall be provided by suitable derating from the part specification values in order to assure high equipment reliability under all applicable service conditions. The design shall include all possible features which will result in reliable and stable operation and hence reduce the requirements for maintenance by highly skilled personnel.

3.4.1 Specified reliability. A reliability program shall be conducted in accordance with the requirements of MIL-STD-785, as appropriate. The switchboard shall have a calculated system mean-time-between-failures of 5,000 hours or better, where a system failure is defined as a failure or failures which disrupt service to four or more line/trunks. Calculations shall reflect a fixed ground environment at 77°F. The same configuration of equipment shall also be required to meet a functional reliability requirement expressed in terms of lost-calls allowed. A lost-call is defined in 4.6.11.3.1. The specified lost-call rate (Po) shall not exceed the following levels:

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Temperature	Po (Lost Call Rate)
$25^{\circ}F \pm 3.6^{\circ}F$	0.305 percent
$70^{\circ}F \pm 3.6^{\circ}F$	0.407 percent
$130^{\circ}F \pm 3.6^{\circ}F$	0.611 percent

These lost-call requirements shall be demonstrated as required by 4.6.11.

3.5 Construction. The switchboard shall be of extremely rugged construction to withstand the environmental exposure and rough handling specified herein. Provisions shall be included in the design and construction to insure mechanical stability and positive positioning of each switchboard in the expanded configuration. The switchboard shall be capable of withstanding the physical stresses as specified in 3.8.

3.5.1 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be physically and functionally interchangeable without modification of such items. Individual items shall not be handpicked for fit or performance.

3.6 Maintainability. A maintainability program shall be conducted in accordance with the requirements of MIL-STD-470, as appropriate. The maintainability program shall encompass the switchboard maintenance concept and requirements contained herein.

3.6.1 Maintenance objective. Maintenance shall be accomplished at the lowest level, as far forward in the battle area as possible. Evacuation to a higher level maintenance organization shall be accomplished only after prescribed fault diagnosis and isolation has been accomplished at the lowest capable authorized echelon.

3.6.2 Maintenance simplicity. Whenever practicable, the switchboard shall contain inexpensive, throwaway parts, standardization and modular design to simplify or enhance maintenance efficiency and operational readiness.

3.6.3 Maintenance personnel. Maintenance personnel skill shall receive maximum use. Maintenance skills and the number of personnel required to maintain equipment shall be minimized by assuring that the switchboard is noncomplex and highly reliable and maintainable.

3.6.4 Maintenance concept.

3.6.4.1 Organizational maintenance. Organizational maintenance of the switchboard shall correct 95 percent of all system-level failures with a mean corrective maintenance time no greater than 15 minutes.

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3.6.4.2 Intermediate maintenance. Faulty lowest replacement items (LRI's) shall be evacuated to intermediate maintenance. Intermediate maintainance shall effect repairs as specified in appropriate techical manuals and as authorized by unit Tables of Organization. Faulty LRI's beyond the capability of intermediate maintenance shall be evacuated to depot.

3.6.4.3 Depot maintenance. Depot maintenance shall include complete repair and rebuild of the switchboard and components.

3.7 Transportability. The switchboard shall be considered a two-man carry load. Carrying handles providing adequate ground clearance shall be provided. The switchboard shall employ, to the maximum extent practical, solid-state and micro-miniaturization technologies, to be transportable as a two-man carry load.

3.7.1 Density. The density of the switchboard in the transport conditions shall be less than that of fresh water.

3.8 Performance characteristics. The switchboard shall be capable of the following performance.

3.8.1 Environmental. The switchboard and the fully loaded transit case shall meet the following environmental conditions as specified herein.

3.8.1.1 Altitude. The switchboard shall operate as specified herein at altitudes up to 10,000 feet above sea level, and the switchboard and transit case shall withstand air transportation at altitudes up to 50,000 feet above sea level.

3.8.1.2 High temperature. The switchboard shall operate as specified herein at ambient air temperature as high as + 130°F. The switchboard, nonoperating, shall withstand exposure to ambient air temperature as high as + 165°F.

3.8.1.3 Low temperature. The switchboard shall operate as specified herein at ambient temperature as low as -25°F. The switchboard, nonoperating, shall withstand exposure to ambient air temperature as low as -65°F.

3.8.1.4 Humidity. The switchboard shall operate as specified herein when subjected to high relative humidities of 100 percent at all temperatures below + 85°F and relative humidities corresponding to 85°F dew point at all temperatures between 85°F and + 120°F and low relative humidities of 5 percent at temperatures of + 120°F. The transit case shall also withstand these humidity conditions.

3.8.1.5 Rain. The switchboard shall operate as specified herein, and shall sustain no leakage, during periods of heavy precipitation with intermittent wind.

3.8.1.6 Immersion. The switchboard and transit case shall be immersion proof to a covering depth of 3 feet of water for a period of not less than 2 hours.

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3.8.1.6.1 Air seal. The switchboard, in its transport condition, and the transit case shall be capable of maintaining a vacuum of 2 pounds per square inch (1 pound per square inch less than atmosphere pressure surrounding the switchboard) with a loss of vacuum no greater than 0.01 pounds during a period of not less than 1 minute. The air-seal test may be performed using the pressure relief valve cutout.

3.8.1.7 Sand and dust. The switchboard, in both operating and nonoperating condition, and the transit case shall withstand exposure to fine sand and dust particles with wind speeds of 35 knots.

3.8.1.8 Salt fog. The switchboard and transit case shall be resistant to the corrosive effects of salt-sea atmosphere.

3.8.1.9 Fungus. The switchboard and transit case in the assembled and ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form or support fungus growth when tested in accordance with Method 508 of MIL-STD-810. Only inherently fungus-resistant grades of material in accordance with Requirement 4 of MIL-STD-454 shall be used. Prior to delivery, all fungus-nutrient contaminations or residues shall be removed.

3.8.1.10 Vibration and shock. The switchboard and transit case shall withstand vibration and shock induced during field transport by military vehicle and by rough handling. The switchboard and transit case shall withstand vibrations encountered during transportation, and the switchboard shall withstand shocks encountered in servicing.

3.8.1.11 Solar radiation. The switchboard shall withstand solar radiation occurring at a rate up to 360 British thermal units per square foot per hour.

3.8.1.12 Static load. The switchboard and transit case shall withstand static load encountered when a 200-pound load is uniformly distributed over a 4-inch by 12-inch area, with covers in place.

3.9 Radio-interference suppression. The susceptibility to interference from external sources and the generation of electromagnetic interference by the equipment shall be controlled in the basic design of the switchboard within the limits of Notice 1 of MIL-STD-461 for class 1C equipment. Design handbook "Interference Reduction Guide for Design Engineers" shall be used as a design guideline to the fullest extent possible.

3.9.1 Bonding and grounds. Bonding for equipment current path return, radio frequency (rf) potential, and shock hazard shall be installed so that expansion, contraction, or movement incident to normal service use will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer, or other high-resistance substances from the immediate area of contact. The equipment finish shall be restored to its original condition in accordance with MIL-F-14072.

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3.9.2 Cables and connectors. Coaxial cables, if used, shall be of the double-shielded type. Other cables requiring shields shall have a minimum of 90 percent coverage. Cable connectors shall be of the approved type, and provided with back shells for fastening the cable shield. The shield shall be constructed of braided, tinned copper wire. All shielded cables shall have the shield terminated in an approved military connector having an rf conductive finish and employing an electromagnetic interference back shell which provides for peripheral bonding of the shield. The use of shielded cable shall be kept to a minimum by the use of alternative suppression techniques such as pin filters. All panel-mounted connectors and component surfaces shall be free of nonconductive finishes, to provide positive bonding and grounding with mating connectors and equipment ground plane.

3.9.3 Emission and susceptibility. The switchboard as exercised through all the modes of operation specified herein shall meet the following emission and susceptibility requirements of MIL-STD-461:

CE01	RE02	CS01	RS03
CE02		CS02	
CE03		CS06	
CE04			

3.10 Dimensions. The dimensions of the switchboard, in the transport configuration, shall not exceed the following:

(a) Height - 12.0 inches

(b) Width - 25.5 inches

(c) Depth - 18 inches

(d) Total volume - 3.2 cubic feet.

3.11 Weight. The weight of the switchboard, in the transport configuration, shall not exceed 66 pounds, including the operator's headset, power cables, and switchboard interconnect cable and hole-in-the-head termination box, excluding the ground rod and cable.

3.12 Color and finish. All fiberglass/plastic materials used for external surfaces shall be impregnated with pigment approximating color green 383 cited in MIL-T-704. All external metal surfaces shall be treated and painted in accordance with MIL-T-704, type G, color green 383.

3.13 Nameplates and marking. Nameplates shall conform to type A of MIL-P-15024. Nomenclature to be applied shall be requested by the contractor in accordance with MIL-E-21981 from the procuring activity. The switchboard shall be serially numbered. Marking shall conform to MIL-M-13231, and front panel markings shall be group I as described therein. Whenever practicable, parts shall be mounted so that their identification markings will be readily visible with minimum or no disassembly of the equipment. All switchboard areas which present a potential safety hazard to personnel shall be clearly marked in accordance with MIL-STD-454, requirement 1.

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3.14 Government-loaned property. The Government shall loan the equipment necessary to perform tests specified in 4.6.7.

3.15 Workmanship. The switchbboard shall be manufactured and assembled in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3).

(b) Quality conformance (see 4.4).

(c) Packaging (see 4.5).

4.3 First article inspection. This inspection shall consist of the inspections specified for Group A, Group B and Group C (see Tables IX, X and XI respectively). Environmental and stress inspections shall be performed in accordance with the order shown in Table XII, according to the number of samples specified therein. This inspection shall not commence until the switchboard has been placed in its final form for delivery.

4.4 Quality conformance. The contractor shall perform the inspections specified in 4.4.1 through 4.4.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all requirements specified herein.

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4.4.1 Group A inspection. Each switchboard shall be inspected for conformance to the inspections specified in Table XI. Discrete lots shall be formed from units which pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105 and using the general inspection levels and acceptable quality levels indicated in Table IX.

4.4.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the procuring activity.

4.4.2 Group B inspection. Group B inspection, including sampling, shall conform to Table X.

4.4.3 Group C inspection. Group C inspection shall be performed on switchboards which have passed Group A inspection. Group C shall consist of the inspections specified in Table XI. Samples shall be selected in accordance with 4.4.3.1. Tests may run concurrently, using separate units.

4.4.3.1 Sampling for Group C inspection.

4.4.3.1.1 Subgroup 1. A sample shall be selected from the first production lot. For subsequent Group C testing, a sample shall be selected every 30 days or 50 production switchboards, whichever occurs first.

4.4.3.1.2 Subgroup 2. A sample shall be selected from the first production lot. For subsequent Group C testing, a sample shall be selected every 120 days or 300 production switchboards, whichever occurs first.

4.4.3.1.3 Subgroup 3. A sample shall be selected from the first production lot. For subsequent Group C testing, a sample shall be selected every 360 days or 500 production switchboards, whichever occurs first.

4.4.3.1.4 Subgroup 4. A sample shall be selected from the first production lot only.

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TABLE IX. Group A inspection.

Inspection or test	Requirement paragraph	Inspection paragraph	Acceptance quality level Major	Minor
Visual/Mechanical	3.2, 3.15	4.6.9.3		
Printed circuit assemblies power module	3.12		1.0%	4.0%
Switchboard combination case	3.7, 3.12		4.0%	10.0%
Switchboard accessories			4.0%	10.0%
Switchboard front panel	3.12, 3.13		4.0%	10.0%
Switchboard rear panel	3.12, 3.13		4.0%	10.0%
Interconnect and power cables			4.0%	10.0%
Air seal	3.8.1.6.1	4.6.10.3.1	1.0%	Not applicable
Electrical inspection			1.0% for the entire electrical inspection	
Basic system operation	3.3.1	4.6.2		
Expansion	3.3	4.6.5		
Terminal/ matrix subsystem	3.3.2.1	4.6.1		
Tone generator and detectors	3.3.2.4			
Operator	3.3.2.3	4.6.3		
Power	3.3.2.6	4.6.4		

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TABLE X. Group B inspection.

Inspection	Requirement paragraph	Inspection paragraph
Interchangeability	3.5.1	4.6.9.2

TABLE XI. Group C inspection.

Inspection or test	Requirement paragraph	Inspection paragraph
Subgroup I		
Immersion	3.8.1.6	4.6.10.3
Reliability	3.4	4.6.11
Subgroup II		
High temperature	3.8.1.2	4.6.10.1
Low temperature	3.8.1.3	4.6.10.2
Transmission	3.3.2.2, 3.3.4	4.6.8
Subgroup III		
Salt fog 1/	3.8.1.8	4.6.10.5
Sand and dust 1/	3.8.1.7	4.6.10.10
Altitude	3.8.1.1	4.6.10.7
Electromagnetic Interference/ electromagnetic compatibility 2/	3.9	4.6.12
Humidity	3.8.1.4	4.6.10.6
Rain	3.8.1.5	4.6.10.4
Vibration and shock	3.8.1.10	4.6.10.9
Subgroup IV		
Fungus	3.8.1.9	4.6.10.8
Weight	3.11	4.6.9.1
Static load	3.8.1.12	4.6.10.12
Sunshine	3.8.1.11	4.6.10.11
Maintainability	3.6	4.6.13
Equipment compatibility	3.3	4.6.7

1/ This switchboard shall be thoroughly washed, cleaned, dried and refurbished after this inspection and before proceeding with subsequent inspection.

2/ The Government reserves the right to require additional testing or to delete any test specified in 4.9. Final determination of reduced testing for Group C inspection shall be made after review by the Government of electromagnetic interference/electromagnetic compatibility test data of the first article.

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TABLE XII. Order of environmental and stress inspection.

Inspection[1/]	ONE	TWO		THREE			FOUR			
		Unit 1	Unit 2	Unit 1	Unit 2	Unit 3	Unit 1	Unit 2	Unit 3	Unit 4
Nondamaging										
Transmission	5	3		1			2			
Temperature (low-high)	6		3		1			1		
Reliability (Temperature)	1	1				1			1	
Reliability (Room)	2	2				2			2	
Potentially damaging										
Bench handling	3		1			3				1
Vibration (subassembly)	4		2			4	1			
Altitude	7	4		2					3	
Dust	8	5			2				4	
Humidity	9	6			3		3			
Fungus[2/]	10	7				5				2
Salt fog	11		4	3				2		
Rain	12		5	4						4
Immersion	13		6	5			4			
Shock	14		7			6				3

1/ Other inspections may precede, follow, or be interspersed between the nondamaging tests, if approved by the Government.

2/ The switchboard shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

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4.4.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in Table XI.

4.4.3.3 Group C failures. Action required relative to Group C failures shall be as specified by the procuring activity.

4.4.4 Reinspection of conforming Group C sample switchboards. Sample switchboards which have been subjected to and passed Group C inspection may be accepted on the contract, provided all damage is repaired and the sample switchboards are resubjected to and pass Group A inspection.

4.4.5 Burn-in. Each switchboard shall be subjected to a 50-hour burn-in at an ambient temperature of 125°F. The power shall be turned on during this period and the switchboard shall be in full operating mode, ready to accept traffic. At the end of the 50-hour period at 125°F, the operational tests specified in 4.6.6 herein shall be performed.

4.4.5.1 Failures. Switchboards which experience a failure during the 50-hour period shall be repaired or adjusted as required, and returned to the burn-in cycle at the 25-hour point, and must complete a 25-hour failure-free period. Failures here are defined in 4.6.11.3.1.

4.4.5.2 Reports of failures. Reports of failures during burn-in shall be handled as specified by the procuring activity.

4.5 Packaging. The completed switchboard, ready for shipment, prior to packaging and packing, shall be examined for completeness, and inclusion of all required items. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.5.1 Rough-handling test. When specified (see 6.2), a rough handling test in accordance with MIL-P-116 shall be performed. The visual, mechanical and functional tests of Group A inspection shall be conducted to determine freedom from operational malfunction caused by the rough handling.

4.6 Test methods.

4.6.1 Terminal and matrix.

4.6.1.1 Termination types. Set up a mixture of termination types throughout the 15 terminal PCB slots, with type III, IV, and V PCB terminations being installed in some of the nine predetermined PCB slots which have been wired for use as 4-wire trunks. There should be a minimum of two of each termination PCB type mode installed for testing, i.e., ringdown line, CBS line, ringdown trunk, rotary dial line, DTMF line, 4-wire converter trunk, DC closure trunk, 6-wire E&M line, 6-wire E&M trunk, and one each of PCB types I, II, III, IV, and V in the unassigned mode. Interconnect the two ringdown trunks in a

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back-to-back connection, also the 4-wire converter trunks and the 6-wire E&M trunks.

4.6.1.1.1 Ringdown line. From a ringdown line termination, originate a call to operator by ringing in. Observe that:

(a) Ring-in is detected by terminal status lamp 1/2-Hz rate.

(b) 500-Hz wait-in-queue tone is heard on line.

(c) Call-answer button is illuminated.

(d) Audible call waiting alarm is sounding.

Operator shall answer by pressing call-answer button. Observe that:

(a) Call waiting lamp is extinguished.

(b) Wait-in-queue tone is disconnected from line.

(c) Call waiting alarm ceases sounding.

(d) Operator and line subscriber have connection.

(e) Terminal status lamp is on fast flash (2 Hz).

Operator shall extend call to ringdown line subscriber by digiting the appropriate 3-digit terminal number on the keyset. Observe that:

(a) Called terminal status lamp illuminates steadily.

(b) Ring back tone is heard in operator headset and calling line.

(c) Two bursts of 20-Hz ring signal are sent to called terminal.

(d) Ring-forward and ringback are in synchronism.

(e) Connection is made between calling and called terminals, and the operator.

(f) Actuation of "Operator Release" disconnects operator, and switches status lamp of calling terminal from fast flash to steady-on.

From either ringdown line terminal ring-off and observe:

(a) Ring-off is detected by terminal status lamp extinguishing.

(b) Associated party terminal-status lamp extinguishes simultaneously.

(c) The connection between parties is broken.

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4.6.1.1.2 Common battery supervision line. From a CBS line termination, originate a call to the operator by going off-hook; have the operator answer the call, and extend call to another CBS line subscriber, making the same observations specified in 4.6.1.1.1.

4.6.1.1.3 Ringdown trunk. From a ringdown line termination, originate call to operator as specified in 4.6.1.1.1. The operator shall call-answer and extend to a ringdown trunk termination while observing that:

(a) Calling party and operator hear one burst of ringback tone in synchronism with ring-forward.

(b) Terminal status lamp associated with trunk illuminates steady-on.

(c) Within approximately 4 s, the ringdown trunk on the other end of back-to-back connection detects 20-Hz ring as evidenced by its status lamp illuminating slow flash.

(d) Calling party and operator hear wait-in-queue tone.

Operator shall press call-answer button to service incoming ringdown trunk call. Observe that:

(a) Call waiting lamp is extinguished.

(b) Status lamp on incoming terminal changes from slow to fast flash.

(c) Wait-in-queue tone is disconnected from original calling party and operator.

(d) Connection is maintained between calling party and outgoing ringdown trunk as evidenced by conversation between operator connection on incoming trunk terminal and original calling party.

Operator shall extend incoming trunk termination by digiting the termination number of another ringdown line subscriber. Observe that:

(a) Called terminal status lamp illuminates steady-on.

(b) Ringback tone is heard by operator and distant calling party.

(c) Two bursts of 20-Hz ring signal are sent to called line, while synchronous ringback tone is heard.

(d) Actuation of operator release button disconnects operator, and switches status lamp of calling trunk from fast flash to steady-on.

(e) Calling and called parties can converse over links via back-to-back ringdown trunk connection.

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From either line subscriber, initiate ring-off and observe that:

(a) Subscriber's status lamp and associated trunk-status lamps are extinguished simultaneously.

(b) Voice connection is broken.

(c) Ring (release) is forwarded from the outgoing trunk and detected by the incoming trunk, as evidenced by extinguishing of the incoming trunk status lamp within 4 s.

(d) The status lamp of the line subscriber extinguishes when the associated incoming trunk lamp extinguishes.

4.6.1.1.4 Rotary dial line. From a rotary dial line, originate a call to another rotary dial line by dialing the appropriate 3-digit number. Observe that:

(a) When going off-hook, the calling party hears dial tone and status lamp is steady-on.

(b) There is no dial-tone time limit.

(c) Upon first digit dialed, dial tone is broken.

(d) Upon third digit dialed, automatic call extension takes place, as evidenced by called-status lamp switching to steady-on condition.

(e) Four bursts of ring-forward are directed toward the called terminal, while ringback tone is heard by the calling terminal.

(f) An overload of the ring generator, caused by the called termination going off-hook during the first burst of ringing, will abort the second burst.

(g) A matrix connection is made and parties may converse. Either party shall go on-hook, while observing that both status lamps extinguish simultaneously.

4.6.1.1.5 Dual-tone multifrequency line. From a DTMF line subscriber, originate a call to another DTMF line subscriber by digiting the appropriate 3-digit number. Observe that:

(a) When going off-hook, the calling party hears dial tone and status lamp illuminates steadily.

(b) Dial tone will time-out after 7-1/2 s, status lamp extinguishes, and busy tone is returned to subscriber line.

(c) Upon receipt of dial tone and first digit dialed, dial tone is broken.

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(d) Upon receipt of third digit, call is automatically extended as evidenced by called terminal status lamp on steady.

(e) Called terminal gets four bursts of 20-Hz ring-forward, while calling terminal receives ringback tone.

(f) When called terminal goes off-hook, there is a connection between parties.

(g) When either party goes on-hook, both status lamps are extinguished simultaneously.

4.6.1.1.6 4-wire tone signaling trunk (converter mode). From a DTMF line subscriber, originate a call to a 4-wire tone-signaling trunk by digiting the appropriate 3-digit number. Observe that:

(a) When going off-hook, the calling party hears dial tone and status lamp illuminates steadily.

(b) Upon receipt of dial tone and first digit dialed, dial tone is broken.

(c) Upon receipt of third digit, call is automatically extended, as evidenced by called trunk-terminal status lamp on steadily when trunk is seized.

(d) Far-end trunk-terminal status lamp switches to steady-on.

(e) At end of acknowledge tone from distant end of trunk, a second dial tone is heard, and if a digit is not dialed within 4 s, error tone is heard by the calling party and the trunk is released from the far end.

(f) When second dial tone commences, another DTMF subscriber line can be called by digiting the appropriate 3-digit terminal number.

(g) Upon receipt of third digit, call is automatically extended as evidenced by called terminal status lamp on steady.

(h) Called terminal gets two bursts of 20-Hz ring-forward, while calling terminal received ringback tone.

(i) When called terminal goes off-hook, there is a connection between parties via the back-to-back trunk terminations.

(j) When either party goes on-hook, the subscriber's status lamp and associated trunk-status lamps extinguish simultaneously.

(k) Voice connection is broken.

(l) Release is forwarded from the outgoing trunk and detected by the incoming trunk, as evidenced by extinguishing of the incoming trunk status lamp within 4 s.

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(m) The status lamp of the line subscriber extinguishes when the associated incoming trunk lamp extinguishes.

4.6.1.1.7 6-wire E&M trunk (DTMF trunk mode, automatic special classmark). From a DTMF line subscriber, originate a call to a 6-wire E&M trunk by digiting the appropriate 3-digit number. Observe that:

(a) When going off-hook, the calling party hears dial tone and the status lamp illuminates steady.

(b) Upon receipt of dial tone and first digit dialed, dial tone is broken.

(c) Upon receipt of third digit, call is automatically extended, as evidenced by called trunk terminal status lamp on steady when trunk is seized.

(d) Far end trunk-terminal status lamp switches on steady.

(e) After the signaling sequence is completed, a second dial tone is heard, and if a digit is not dialed within 15 s, busy tone is heard by the calling party and the trunk is released from the far end.

(f) When the second tone commences, another DTMF subscriber line can be called by digiting the appropriate 3-digit terminal number.

(g) Upon receipt of the third digit, call is automatically extended as evidenced by called terminal status lamp on steady.

(h) Called terminal gets bursts of 20-Hz ring-forward, while calling terminal receives ringback tone.

(i) When called terminal goes off-hook, there is a connection between parties via back-to-back trunk terminations.

(j) When either party goes on-hook, the subscriber's status lamp and associated trunk status extinguish simultaneously.

(k) Voice connection is broken.

(l) Release is forwarded from the outgoing trunk and detected by the incoming trunk, as evidenced by extinguishing of the incoming trunk status lamp within 4 s.

(m) The status lamp of the line subscriber extinguishes when the associated incoming trunk lamp extinguishes.

4.6.1.1.8 2-wire DC closure trunk (semiautomatic) DTMF mode. From a DTMF line subscriber, originate a call to a 2-wire DC closure trunk by digiting the appropriate 3-digit number. Observe that:

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(a) When going off-hook, the calling party hears dial tone and status lamp illuminates steadily.

(b) Upon receipt of dial tone and first digit dialed, dial tone is broken.

(c) Upon receipt of third digit, call is automatically extended, as evidenced by called trunk terminal status lamp on steady when trunk is seized.

(d) A second dial tone is heard, and if a digit is not dialed within 4 s, error tone is heard by the calling party and the trunk is released.

(e) When second dial tone commences, a Dial Central Office (DCO) subscriber line can be called by digiting the appropriate terminal number, using the DCO's numbering plan.

(f) Upon receipt of digits, call is extended in the fashion of the DCO called.

(g) Called terminal gets bursts of ring-forward, while the calling terminal receives ringback tone.

(h) When called terminal goes off-hook, there is a voice connection between parties via the trunk termination.

(i) When the switchboard party goes on-hook, the subscriber's status lamp and associated trunk status lamp extinguish simultaneously.

(j) Voice connection is broken.

(k) Release is forwarded from the outgoing trunk and detected by the DCO, as evidenced by extinguishing of the outgoing trunk status lamp.

4.6.1.1.9 Unassigned terminations. Place one of the type I, II, III, IV, or V termination PCB's in the unassigned mode of operation. With a DTMF line termination, place a call to each termination type in the unassigned mode, digiting the appropriate 3-digit termination numbers, and observe that:

(a) After the third digit is dialed, the calling terminal status lamp switches to slow flash and wait-in-queue tone is heard by the calling party.

(b) The call waiting lamp is illuminated under the call-answer button to indicate the call waiting-in-queue.

(c) The audible call waiting alarm sounds until the operator actuates the call-answer button.

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Operator shall press call-answer to service incoming line subsriber who misdialed the unassigned termination. Observe that:

(a) Call waiting lamp is extinguished.

(b) Status lamp in incoming terminal changes from slow to fast flash.

(c) Wait-in-queue tone is disconnected from calling party.

(d) Operator and calling party have connection.

Operator shall press operator release button to disconnect from calling party. Observe that:

(a) Voice connection is broken.

(b) Operator call status lamp and terminal status lamp are extinguished.

Repeating the above procedure, demonstrate that the line subscriber dialing of unassigned terminations of the other type of PCB's also produces routing to the operator's queue as a switchboard response.

4.6.1.2 Matrix subsystem.

4.6.1.2.1 Crosspoint matrix (30-line board). Utilizing the mix of termination types specified in 4.6.1.1, but with all terminals assigned, establish 15 simultaneous calls in the single configuration, noting the following in each call:

(a) That the calling party receives the proper switchboard response upon going off-hook/ringing-in (dial-tone or wait-in-queue tone) according to terminal type.

(b) That the switchboard collects the signaled digits from the subscriber (or operator) when calls are extended.

(c) The ringback tone is heard when call is extended to line subscriber and that subset rings.

(d) The terminal status lamps always reflect the proper idle-busy states as seen in 4.6.1.1.

(e) That after called terminal answers, a crosspoint matrix connection has been established.

After all 15 calls have been established, reconfirm that all 30 terminal-status lamps are on steady. Verify all path connections by conversing over each one without changing any terminal's supervision state. Verify that a 16th link is not available, by placing a call from the operator to test tone facility. Observe that:

(a) Operator upon dialing 199 receives busy tone.

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(b) All 30 status lamps are on steady.

4.6.1.2.2 Crosspoint matrix (60-line board). Interconnect two 30-line switchboards to form a 60-line stack with number plan switches on the rear panels set for 1 to 30 on lower board and 31 to 60 on upper board. Place one master/slave switch in master mode and the other in slave mode. Utilizing all type II PCB terminations for convenience in the second switchboard, place 30 consecutive calls to the 30 terminations in the lower board, making the same supervision, service tone, and status lamp observations as in the single switchboard test. Verify that a 31st link is not available by having the operator dial test tone and receive busy tone instead.

4.6.1.2.3 Crosspoint matrix (90-line board). Add a third switchboard to the stack, with number plan switch in the 61 to 90 position, and master/slave toggle in the slave position. Utilizing all type II PCB terminations for convenience in the third switchboard, place calls as specified in 4.6.1.2.2 until the first 60 terminals are busy. Place a 31st call, using terminal number 61, and observe:

(a) That the calling subscriber receives busy tone.

(b) That the calling-status lamp remains off.

Replace subset on-hook, and also that of terminal 60. Observe that a call may not be placed from terminal 61 by:

(a) Receiving dial tone when going off-hook on terminal 61.

(b) Dialing terminal 60, causing that terminal to ring and sending ringback tone to calling party.

(c) Observing terminal 60 status lamp lights steadily when phone is rung.

(d) Noting, upon terminal 60 answering, that there is an established path.

By replacing subsets 59 through 46, links shall become available to the third switchboard for call extension from terminals 62 through 75. These calls shall then be set up sequentially while observing the appropriate supervision, idle-busy lamp status, and service tone occurrences, while placing each call until all links are again used. Finally place call from terminal 76 dialing 47, and observe that 76 receives busy tone.

4.6.1.2.4 Service tone matrix (30-line board).

4.6.1.2.4.1 Subscriber service tones. Utilizing a DTMF line subset, establish a call to the operator digiting 0. Observe that calling subset can hear wait-in-queue tone (500 Hz), 2 s on, 2 s off, continuously. Utilizing a rotary dial line, establish that dial tone can be fed to this type of subset by going off-hook and observing that calling subscriber can hear continuous 500-Hz dial tone for an

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unlimited amount of time, until a first digit is dialed. Establish a call between two local subscribers, and then, with a third subset of the DTMF line type, digit the terminal number of one of the parties already on the established call. Observe that:

(a) The calling party after keying third digit receives busy tone, 500-Hz, 0.5 s on 0.5 s off.

(b) When original two-party call was placed as 20-Hz ring-forward was occurring, ringback tone of 500 Hz, 2 s on, 2 s off was heard.

On the established two-party call above, hang up one phone, leaving the other off-hook idle-busy. Via operator keyset, digit the number of the idle-busy phone and observe that:

(a) Operator and called party receive off-hook warning tone (3125 Hz), 0.5 s on, 0.5 s off.

(b) When operator releases, off-hook warning tone is removed from the connection, or

(c) When called party goes on-hook, off-hook warning tone is removed from the connection.

Establish three calls simultaneously from DTMF lines, to verify that three DTMF receivers are available via matrix connection for the placement of three simultaneous call extensions in each 30-line switchboard. Observe that:

(a) When lines are off-hook, they are immediately fed dial tone (and not busy tone).

(b) When 3rd digit is dialed, automatic call extension takes place as evidenced by the ringing of three local subscribers.

4.6.1.2.4.2 Operator service tones. From the front panel, the operator shall establish a call to a line subscriber to verify that the operator terminal may receive service tones via the service-tone matrix. Observe that after third digit is dialed, ringback tone (500 Hz), 2 s on, 2 s off, may be heard in the operator's headset. From the front panel, operator shall assist a ringdown caller in operator queue by call answer and call extension to a busy terminal. Observe that:

(a) Busy tone (500 Hz), 0.5 s on, 0.5 s off, is heard in the operator's headset.

(b) When called release button is actuated, busy tone is removed.

From the front panel, operator shall digit the number of a terminal which is outside the numbering plan. Observe that error tone (1 kHz), 0.25 s on, 0.25 s off, is heard in the operator's headset. From the front panel, operator shall digit the number of a terminal of the CB

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or CBS type which has been left off-hook when associated party on a call has been placed on hook. Observe that off-hook warning tone (3125 Hz), 0.5 s on, 0.5 s off, is continuously heard on the operator's headset.

4.6.2 Basic system (30-line). Set up two switchboards with an identical mixture of termination types throughout 15-terminal PCB slots with a fairly even distribution of subset types for the first eight slots, with the last seven slots being a mix of trunks as shown in Table XIII. All other card slots are assumed filled with the basic PCB set plus automatic operation kit.

4.6.2.1 Number plan dialing and directory look-up.

4.6.2.1.1 Number plan selectability. With number plan rear-panel switch in each of its three positions, dial a termination within the selected number group from a type II PCB subset, and observe that automatic call extension takes place to the assigned termination. In each position of number plan switch, originate a call to a termination outside the designated number plan and observe that the call is routed to operator queue. Exclusive of number plan switch position, originate a call to a 7xx and an 8xx number and observe that the call is routed to operator queue. Also observe that, when an unassigned termination is dialed, the call is routed to operator queue.

4.6.2.1.2 Directory look-up operations. Table XIV lists all possible calling and called terminations in calls involved in 9xx type of call extension.

4.6.2.1.2.1 Automatic terminations to manual trunks. For each called trunk possibility listed in Table XIV, Section A, program the trunk directory memory with the switch locator assignments from Table XIII. Originate calls as listed in Section A and dial the switch locator assigned to the called terminal. Observe that:

(a) Call is extended to the 9 switch location (SL) associated terminations. Observe the associated terminal status lamp switches to steady-on state.

(b) Seize is forwarded to the terminating switchboard terminal as evidenced by the indicators of that switchboard.

(c) Call is extended to the manual switchboard subscriber, using that board's manual procedures.

(d) A voice path exists between distant parties.

(e) Call can be broken down by either release from the manual switchboard subscriber or the original calling party going on-hook.

(f) Call link has been broken by the indication of both terminal-status lamps extinguishing simultaneously.

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TABLE XIII. Trunk types.

SL	TERM #	TERM TYPE	ATTACHED TO	ASSIGNED	SPECIAL CLASSMARK
60	X30	4w Conv	CV-1919	YES	DID
60	X29	4w Conv	CV-1919	YES	DID
	X28	CB DTMF	TA-938	YES	
61	X27	E&M dp Trk	SB-3614	YES	AUTO
	X26	CB DTMF	TA-938	YES	
62	X25	E&M DTMF Trk	SB-3614	YES	AUTO
	X24	DB dp	TA-236	YES	
	X23	E&M dp line	TBD	YES	
	X22	CB dp	TA-236	YES	
	X21	E&M DTMF line	TBD	YES	
65	X20	PABX	AN/TTC-38	YES	
65	X19	PABX	AN/TTC-38	YES	
70	X18	DC closure dual	DCO	YES	DOD
70	X17	DC closure dual	DCO	YES	DOD
75	X16	4w Conv	SB-3614	YES	
75	X15	4w Conv	SB-3614	YES	
80	X14	DC closure dp	DCO	YES	DOD
80	X13	DC closure dp	DCO	TES	
	X12	CB dp	TA-236	YES	
	X11	CB DTMF	TA-938	YES	
	X10	Ringdown line	TA-312/977	YES	DID
	X09	CBS line	TA-312	YES	
85	X08	Ringdown Trk	SB-3614	YES	AUTO
85	X07	Ringdown Trk	SB-3614	YES	AUTO
90	X06	Ringdown Trk	SB-22	YES	
95	X05	Ringdown Trk	SB-22/955	YES	DID
	X04	CB dp	TA-236	YES	
	X03	CB DTMF	TA-838	YES	
	X02	CBS line	TA-312	YES	
	X01	Ringdown line	TA-312	YES	

TABLE XIV. Directory look-up termination possibilities.

CALLING	CALLED
SECTION A (any of the following)	
Ringdown Trunk DID	Ringdown Trunk
Ringdown Trunk AUTO	4w Conv DID
CB DTMF line	6w E&M Trunk
CB dp line	6w E&M Trunk DID DTMF
4w Conv	6w E&M Trunk DID dp
4w Conv DID	DC Closure Trunk
6w E&M Trunk AUTO DTMF	Ringdown Trunk DID
6w E&M Trunk AUTO dp	
6w E&M Line DTMF	
6w E&M Line dp	
6w E&M Trunk DID DTMF	
6w E&M Trunk DID dp	
Ringdown Line DTMF	

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TABLE XIV. Directory look-up termination possibilities. (Cont'd)

CALLING	CALLED
SECTION B (any of the following)	
Ringdown Trunk DID	Ringdown Trunk AUTO
Ringdown Trunk AUTO	4w Conv
CB DTMF line	4w PABX
4w conv	6w E&M Trunk AUTO DTMF
4w conv DID	6w E&M Trunk AUTO dp
6w E&M Trunk AUTO DTMF	6w E&M Trunk DID DTMF
6w E&M Trunk DID DTMF	6w E&M Trunk DOD dp
6w E&M Line DTMF	DC Closure Trunk DOD dual
Ringdown Line DTMF	DC Closure Trunk DOD dp
SECTION C (any of the following)	
CB dp Line	6w E&M Trunk AUTO dp
6w E&M Trunk AUTO dp	6w E&M Trunk DOD dp
6w E&M Line dp	DC Closure Trunk DOD dp
6w E&M Trunk DID dp	DC Closure Trunk DOD dual
SECTION D	
4w PABX	Ringdown Trunk AUTO 4w Conv 6w E&M Trunk AUTO DTMF 6w E&M Trunk AUTO dp 6w E&M Trunk DOD DTMF 6w E&M Trunk DOD dp DC Closure Trunk DOD DTMF DC Closure Trunk DOD dp
SECTION E	
4w PABX	Ringdown Trunk 4w Conv DID 6w E&M Trunk 6w E&M Trunk DID DTMF 6w E&M Trunk DID dp DC Closure Trunk Ringdown Trunk DID

NOTES: DID- direct inward dialing, manual outgoing
DOD - direct outward dialing, manual incoming
AUTO- direct inward dialing, direct outward dialing
dp- Rotary dial signaling
DTMF- dual-tone (key) signaling
Unless otherwise noted all signaling is in the default mode of the partiular PCB type

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4.6.2.1.2.2 Dual-tone termination to automatic trunk terminations. Originate the calls listed in Table XIV Section B from a DTMF termination by dialing the 9 switch locators assigned to the automatic trunks. Observe that:

(a) Call is extended to the 9 switch locator associated terminations and its terminal status lamp switches to steady-on state.

(b) Second dial tone (425 Hz) is received from the called automatic switchboard.

(c) Call may now be extended directly to a local subscriber using that office's numbering plan.

(d) When called party answers, a voice connection has been established.

(e) Call can be broken down by either subscriber going on-hook.

(f) Call link in the switchboard has been broken, as indicated by the calling and outgoing trunk terminal status lamps switching off simultaneously upon receipt of release.

4.6.2.1.2.3 Rotary dial termination to automatic trunk terminations. Originate the calls listed in Table XIV Section C from the rotary dial terminations, dialing the 9 switch locators assigned to the called trunks (Table XIII). Observe that:

(a) Call is extended to the 9 switch locator associated terminal, by its terminal status lamp switching to steady-on state.

(b) Second dial tone is received from the other switchboard.

(c) Call may now be extended directly to any local subscriber by using that office's numbering plan.

(d) When called party answers, a voice connection has been established.

(e) Call can be broken down by either party going on-hook.

4.6.2.1.2.4 4-Wire pabx trunk to automatic trunks. Originate the calls listed in Table XIV Section C over a 4-wire pabx trunk from an AN/TTC-38 local subscriber within the PR-SL-XXX numbering plan of that office. Use a 9 switch locator as the last three digits dialed which is associated with one assigned in the switchboard to the automatic trunks to other switch-boards. Observe that:

(a) Call is extended to the 9 switch locator associated terminal, by its terminal status lamp switching to steady-on state.

(b) Second dial tone is received from the other switchboard.

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(c) Call may now be extended directly to any local subscriber by using that switchboard's numbering plan.

(d) When called party answers, a voice connection has been established.

(e) Call can be broken down by either party going on-hook.

(f) The call link of the switchboard has been broken, as indicated by outgoing and incoming trunk terminal-status lamps switching off.

4.6.2.1.2.5 4-Wire pabx trunk to manual trunks. Originate the calls listed in Table XIV Section D over a 4-wire pabx trunk from an AN/TTC-38 local subscriber within the PR-SL-XXX number plan of that office, using 9 switch locator as the last 3 digits dialed. The 9 switch locator dialed is one assigned in the switchboard to a manually outgoing trunk to a switchboard or adapter. Observe that:

(a) After the AN/TTC-38 subscriber dials PR-SL-9SL, call is extended to the 9 switch locator associated terminal, by its terminal-status lamp switching to steady-on state.

(b) Seize is forwarded to the manual terminal, as evidenced by the indicators of the called equipment.

(c) Call is extended to a subscriber, using the called equipments manual procedures.

(d) A voice path exists between distant parties.

(e) Call can be broken down by either release from the manual equipment or the original calling party going on-hook.

(f) Call link has been broken by the indication of both trunk terminal status lamps extinguishing simultaneously.

4.6.2.1.3 Illegal subset dialing. The following are cases of inappropriate dialing from the type of subset or incoming trunk. In each case, verify that the call is routed to local operator queue, as evidenced by calling party hearing wait-in-queue tone and audible call-waiting alarm on front panel:

(a) Rotary dial subscriber or trunk dials the terminal number of a 2-wire ringdown trunk bearing the special DTMF trunk classmark.

(b) Rotary dial subscriber or trunk dials the terminal number of a 4-wire pabx trunk.

(c) Rotary dial subscriber dials the terminal number of a 4-wire converter trunk.

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(d) 4-wire pabx trunk dials a terminal number of a 4-wire pabx trunk.

(e) Rotary dial subscriber or trunk dials the terminal number of a 6-wire E&M DTMF trunk bearing the special automatic classmark.

4.6.2.1.4 Dialing to malfunctioning trunk. Remove the transmit or receive pair from the rear-panel termination-box connection associated with the CV-1919 trunk connection. A DTMF line subscriber shall dial the 9 SL associated with that trunk, or shall dial the actual termination number. Observe that, after seize time-out, the call is extended over another trunk in the SL group, and the malfunctioning trunk shall display an out-of-service indicator.

4.6.2.2 Standard call features.

4.6.2.2.1 Operator queuing of subsets. Program the trunk directory memory with priority precedence classmark for one of each of the four subset types. Establish for each subset, one routine and one priority call and determine that the call is routed to the appropriate precedence level operator queue. In the priority authorization case, observe the following:

(a) For ringdown line, ring-in produces interrupted audible night alarm and call-answer lamp flash.

(b) For CBS line, off-hook produces the same effect.

(c) For rotary dial, dialing 0 produces the same effect.

(d) For DTMF line, dialing R0 produces the same effect.

In the routine authorization cases, observe the following:

(a) For ringdown line, ring-in produces steady audible night alarm and call-answer lamp.

(b) For CBS line, off-hook produces same effect.

(c) For rotary dial, dialing 0 produces same effect.

(d) For DTMF line, dialing 0 produces same effect.

Observe that in the DTMF line case a routinely classmarked line which dials R0 gets error tone, and that a priority class line gets routine operator queue when dialing 0.

4.6.2.2.2 First-in-first-out queuing and queuing capacity. Verify, by use of a mix of subset types, that routinely classed subsets are processed through operator queue on a first-in-first-out basis. In each case, observe that:

(a) The subset receives wait-in-queue tone when dialing 0/ring-in/off-hook appropriate to subset.

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(b) The audible alarm and call-answer are on steady.

(c) The audible alarm is switched off automatically when operator presses call-answer button.

(d) The wait-in-queue tone ceases to subset when operator services the subset.

(e) The order in which subsets are connected to the operator when call-answer button is actuated is in the same order as the calls were placed.

Make the same verification for all priority class subsets. Then, with a mix of routine and priority subsets, intermix the sequence with which calls are placed and verify that all priority class subsets are processed on a first-in-first-out basis, and then all routine subsets are processed on a first-in-first-out basis as call-answer is actuated. Verify that the operator queue will hold 15 callers by placing calls to queue from 15 subsets, while getting the appropriate subset and switchboard responses to queuing. Verify that the 16th subset which calls for queue receives busy tone. Intersperse recall attempts randomly in the mix of routine and priority queued calls. Verify that recall precedes routine and is after priority queued calls.

4.6.2.2.3 Call extension and line preemption. Establish a routinely marked 2-party call between two line subscribers. Verify for the routine and priority classed DTMF subset that when the terminal number of a busy party in the above 2-party call is dialed, busy tone is received and the calling party's status lamp does not light. Verify for the priority classed subset that, when R is dialed ahead of the busy terminal's number, preemption automatically takes place. Observe that preemption is heard by all three parties and at the end of the tones a new connection is established between new calling party and the terminal dialed.

4.6.2.3 Special call features.

4.6.2.3.1 Operator recall. Establish a two-party call between two type II PCB subsets. After call is set up, one subscriber shall actuate hook-switch to recall the operator to the connection. Observe that the audible alarm and the call-answer lamp switches are on and that actuation of the call-answer button connects the operator to the recalling terminal. Note that terminal status lamp switches to fast flash. Verify that the operator may disconnect from the call by actuation of the operator release button.

4.6.2.3.2 Operator intercept. Classmark a subset for operator intercept feature. Utilizing a line subset, originate a call to the classmarked subset and observe after the third digit that the calling subscriber is routed to operator queue.

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4.6.2.4 Trunk operations.

4.6.2.4.1 Trunk hunting with directory. Establish a call between a line subscriber and a Switchboard SB-22 trunk, by dialing the highest numbered terminal number of the group. Using another line subscriber's subset, dial the 9 switch locator associated with that group and verify that the call is extended to the next lower number in the group. All terminations in the trunk group now being busy, a third call origination should receive a busy tone.

4.6.2.4.2 Trunk hunting without directory. With the trunk directory memory removed from the switchboard, establish a call to the highest numbered terminal in the trunk group to the Switchboard SB-22. Activate trunk hunting control No. 1. Using the methods outlined in 4.6.2.4.1, establish calls to the trunk group, each time dialing the same termination number while noting that the calls are being established on consecutively numbered terinals to the Switchboard SB-22 until the group is filled. One more call placement beyond the filled group shall produce busy tone. Repeat this process with trunk hunting control No. 2 and the group of terminals associated with the CV-1919 trunks.

4.6.2.4.3 Precedence monitor. Precedence monitoring shall be tested for the connections shown in Table XV (any calling to any called). Utilizing the call extension and directory look-up operations outlined in 4.6.2.1.2, cover all possible calling and called party combinations listed in the Table XIV. Through the use of interswitchboard trunk connections as shown in Table XIII, demonstrate that the calling party precedence authorization is granted or disallowed (by releasing connection) when initial outgoing trunk is dialed. Demonstrate also that subsequent tandem trunk connections are made based on consistency of dialing on both incoming nd outgoing dialing (i.e. R9SL-R9SL or 9SL-9SL), and that when a final call extension is made to a 4-wire pabx trunk to a AN/TTC-38, precedence levels higher than R1 or P are released.

TABLE XV. Precedence monitoring connections.

Calling	Called
2-wire DTMF line	4-wire converter trunk
2-wire DTMF trunk	4-wire pabx trunk
4-wire converter trunk	2-wire DTMF trunk
6-wire E&M DTMF Trunk	6-wire E&M Trunk DTMF
6-wire E&M DTMF Line	2-wire DC Closure trunk

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4.6.2.4.4 Trunk preemption. Establish a line-to-trunk call routinely on a Switchboard SB-22 trunk. Using a precedence classmarked DTMF line subscriber, dial Rxxx utilizing the xxx associated with one of the Switchboard SB-22 trunks. Observe that:

(a) Preemption tone is heard by all parties.

(b) The called termination is released toward the Switchboard SB-22 (as evidenced by catseye drop).

(c) Connection is reestablished between preemptor and Switchboard SB-22 terminal.

Establish a line-to-trunk call routinely on a 4-wire converter trunk (back-to-back connection) to another switchboard, and on second dial-tone, dial routinely to a subset. Using a precedence classmarked DTMF line subscriber, dial Rxxx utilizing the xxx associated with that of the outgoing 4-wire converter trunk. Observe that:

(a) Preempt tone is heard by all.

(b) The outgoing trunk is released.

(c) The status lamps of outgoing trunk, incoming trunk and original distant subscriber turn off.

(d) The outgoing trunk is reseized.

(e) Dial tone is returned to calling party.

(f) Call extension is now possible to another line subscriber at distant end of trunk (dial Rxxx).

(g) Status lamps of reextended call switch on steady.

(h) Once the priority call is established, it is not again preemptable from a priority class subset even when dialing R.

The above process shall be repeated for:

4-wire pabx trunk to AN/TTC-38
2-wire DC closure trunk to a DCO
6-wire E&M trunk to a SB-3614/TT

4.6.2.4.5 Trunk routining. Trunk routining shall be tested for all type III signaling modes. Set up one trunk in each signaling mode and verify its operation. Disconnect one signal wire from the terminal box for each trunk and observe that the terminal status lamp shall flash at a 1 Hz rate after a maximum wait of 120 s. Reconnect the signal wires and observe that the terminal status lamps return to the idle (off) state after a maximum of an additional 120 s. Disconnect the signal wire of one trunk and attempt to seize that trunk from the operator's console. Observe that after the call attempt, the terminal status lamp shall flash at a 1 Hz rate.

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Reconnect the wire and attempt the call again, observe that after the call completion the terminal status lamp shall return to the idle (off) state. Repeat this procedure for an incoming call attempt and for the other signaling modes.

4.6.3 Operator subsystem.

4.6.3.1 Call-answer. Call-waiting lamp and call-answer function may be demonstrated as part of tests specified in 4.6.2.2.1 and 4.6.2.2.2.

4.6.3.2 Call extension. Originate a call to operator by going off-hook with a CBS subset (type I PCB termination). Switchboard reactions shall be observed as specified in 4.6.2.2.1. After operator activates call-answer button, the call shall be extended using the 4x4 keyset on front panel by dialing the terminal number of an assigned line subscriber. Observe that:

(a) Operator and calling party hear ringback tone synchronously with ring-forward to called termination.

(b) Call-status lamp and called-terminal status lamps illuminate steadily.

(c) When called terminal answers, there is a voice connection.

(d) Operator activation of either call-answer or operator-release button switches calling status lamp from fast flash to steady on.

(e) The operator is disconnected from the voice connection.

4.6.3.3 Operator preemption. Establish a line-to-line call as specified in 4.6.2.2.3. Originate a call as specified in 4.6.3.2 to operator. The operator shall take the role of calling party in extending the call as specified in 4.6.2.2.3. Observe the same switchboard reactions to operator preemption as occurred in the line preemption case. (Operator is always assumed to have priority authorization.) Additionally, observe in each step that the call-status lamp takes on an illumination mode consistent with priority authorization level of unextended party or precedence level of call link when extended.

4.6.3.4 Call monitoring. Observe that when the operator digits the terminal number of a terminal engaged in a call, the operator is connected into the existing call. Observe also that no phones ring, that the call-status lamp takes on the illumination mode of the precedence level of the call, and that the digited terminal's status lamp switches to fast flash. Operator-release button shall be activated to release operator from call. Observe lack of operator connection and that fast flashing lamp has returned to steady mode. Also observe that call-status lamp has switched off. Reestablish the above call, and verify that operation of the call-release button disconnects the party not dialed by the operator. Observe that the

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disconnected party's status lamp switches off, and that he has no voice connection. Demonstrate that the party still connected to the operator may now be extended to another party using the 4x4 keyset on the front panel. Observe the usual tone and lamp responses of the switchboard during the call extension mode.

4.6.3.5 Call holding. As part of the procedures specified in 4.6.2.2.1, place an unextended call on hold by pressing the reentry button. Observe that:

(a) The lamp under the call-answer button is illuminated as if a priority terminal was on hold.

(b) The call-status lamp is extinguished.

(c) The operator circuit is no longer connected to subscriber, and the sudible alarm sounds while operator circuit is idle.

(d) Another two-party call may be placed by the operator, observing all tone, lamp, and connection responses of the switchboard.

(e) Pressing the call-answer button retrieves the connection between the on-hold party and the operator.

(f) The lamp under the call-answer button changes to steady, provided no priority callers are waiting in-queue.

(g) The call status lamp reflects the precedence authorization level of the calling party.

(h) The call may now be extended under the normal rules of call extension while observing all switchboard tone, lamp, and connection responses.

4.6.3.6 Reentry. Establish a two-party call between two line subscribers of any type via operator call-extension procedures. As the last step in the procedure, activate the reentry button. Observe that:

(a) The lamp under the reentry button illuminates.

(b) The operator is disconnected from the two-party call.

(c) The call-status lamp extinguishes.

(d) If there are calls waiting in queue, the audible alarm sounds.

Originate a two-party call by operator's keyset or call extend another call from queue. Then press reentry button while observing that:

(a) Reentry lamp extinguishes.

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(b) Original calling party terminal-status lamp switches to fast flash.

(c) Operator has voice connection to the existing call he has just reentered.

(d) Activation of the operator-release button will disconnect operator circuit from the two-party call.

(e) Calling party status lamp goes back to steady-on condition.

4.6.3.7 Conference. Originate a call to the operator from any line subscriber. The operator shall answer the caller in queue, using the normal call answering procedures. A conference call extension shall be set up, using the rules of normal call extension, except that before digiting the first 3-digit termination number on the 4x4 keyset, the C-key shall be pushed. Each subsequent three digits should result in a call extension to that termination, making a voice connection to the same conference link. At each call extension to a conferee, observe that:

(a) The called terminal-status lamp switches to steady-on.

(b) Synchronous ringback tone (with ring-forward) is heard by the calling party, operator, and all existing conferees.

(c) The conference link will support a ten-party conference.

(d) Call extension to an eleventh party will result in error tone.

(e) Origination of a second several-party conference and subsequent attempt at establishment of a third conference will result in error tone to the operator.

4.6.3.8 Memory entry operations. By utilizing the appropriate memory entry procedures from the front panel, install the following classmark/ trunk directory groups and observe the associated switchboard reactions:

(a) For a given DTMF line termination, observe the line subscriber precedence level reaction upon entry into operator queue for all combinations of dialed precedence versus installed precedence authorization (4.6.2.2.1). Include entry, verify, and delete functions.

(b) For a given line termination of any type, observe the switchboard reaction to the placement of a call from another line termination, when the given termination does and does not have the operator intercept classmark (4.6.2.3.2). Include entry, verify, and delete functions.

(c) Install a precedence classmark on a DTMF line subscriber and also a special converter trunk classmark against the trunk to a CV-1919. Place a call from the DTMF line to the CV-1919 trunk by dialing Rxxx, and reach

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operator queue of the converter. After conversing, the two parties shall release, and the switchboard operator shall delete the special classmark against the 4-wire trunk. Again, place a call from the same DTMF line subset, by dialing Rxxx with xxx being the termination number of the same CV-1919 trunk. Observe that, within a few seconds of seizure of the 4-wire trunk, the lack of dialing an Rxxx shall cause release procedures to take place which break down the call link and release the converter. This switchboard reaction shall verify that precedence monitoring will take place in the absence of a special converter trunk classmark, and that the classmark was successfully installed.

(d) Demonstrate the entry procedures for trunk directory groups as specified in 4.6.2.4.1. As part of this procedure, it shall be shown that deletion of a terminal from switch locator grouping will result in the dropping of of this termination from the trunk-hunting feature.

(e) Demonstrate that erase trunk shall delete all groups which have been installed in the directory, by placing multiple call extensions to these 9 SL numbers and verifying in each case that a second call extension to a 9 SL results in busy tone. Demonstrate that individual delete trunk operations affect the deletion of only those individual trunk groups.

(f) After the installation of a terminal in a trunk group, demonstrate the verify trunk fundtion by dialing in the interrogation of the particular terminal and switch locator number, and observing the busy tone in the operator headset. Also, verify that association of the wrong pair of terminal switch locator numbers shall result in error tone to the operator headset.

4.6.3.9 Lamp test/night light test. Observe that, when the lamp-test switch is activated towards the momentary-on position, all lamps on the switchboard front panel illuminate to the level set by the brightness potentiometer. Also observe that, when the same switch is activated towards the night-light position, all night lamps are fully illuminated and are so oriented as to illuminate the assigned areas of the front panel and writing surfaces.

4.6.3.10 Common control unit alarm test. By the creation of a false common-control failure, show that the common control unit failure lamp illuminates. This false failure may be introduced by placing the master/ slave switch in the slave position on the rear panel of a switchboard formerly assigned as master (in a 30-, 60-, or 90-line stack).

4.6.3.11 20-Hz failure test. By the creation of a false 20-Hz ringer failure, show that the 20-Hz failure lamp illuminates. This false failure may be created by appropriate overloading of the 20-Hz generator terminations.

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4.6.3.12 Night alarm volume control. Demonstrate that the audible alarm volume is variable by the front-panel control from full volume to minimum audible sound level while a caller is waiting in operator queue.

4.6.3.13 Headset volume control. Demonstrate that the operator-circuit volume control has adequate range of adjustment to sufficiently raise the sound level of a trunk circuit to be heard by a person of normal hearing ability with the H182 headset.

4.6.4 Power subsystem. The switchboard shall be configured so that one prime variable power source is connected to the prime power input and two batteries are connected independently to their separate switchboard inputs. The battery's voltages shall be no greater than 25 VDC each. Adjust the prime voltage to 30 VDC and extend four random calls. Observe their successful completion. Adjust the voltage to 24 VDC. Configure the switchboard for maximum load and measure the input current. Observe that calls extended above remain extended. Perform the following adjustments and observations:

(a) With the batteries disconnected, adjust the prime voltage to 21.5 VDC and monitor the flashing DC failure lamp. Further adjust the prime voltage to 19 VDC and monitor the DC failure lamp. The lamp should illuminate steadily.

(b) Return the voltage to 24 VDC and reconnect the batteries. Again, reduce the variable voltage to 19 VDC and monitor the DC alarm. Calls established while the prime voltage was at 24 VDC shall remain established after the voltage is reduced to 19 VDC.

(c) While the switchboard is operating on one battery and with calls established, disconnect the unused battery and connect up a fresh battery. Switch the load of the switchboard to the fresh battery and monitor the status os the switchboard and extended calls.

(d) With the switchboard operating in a normal fashion and with calls being established, monitor the DC failure-alarm operation while the switchboard is connected to battery power only.

(e) Reverse the polarity of the prime voltage to the switchboard. Set the voltage at 26 VDC and turn on the supply. Return the voltage to zero, reverse the voltage input to a correct connection, replace fuse and again set the voltage to 26 VDC. Extend four random calls and observe their successful completion. Repeat this procedure utilizing a battery.

(f) Adjust the prime voltage to 30 VDC $\pm$ 15 percent. Turn on the switchboard. Reduce the voltage to 26 VDC, extend four random calls and observe their successful completion.

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4.6.5 Expansion. The following demonstrations are intended for use with 60- and 90-line stack configurations.

4.6.5.1 Number plan switches. With number-plan switches on rear panels in different and unique combinations, show that call extensions from one line subscriber reach three different terminations during call extension while dialing the same number.

4.6.5.2 Assignment of master switchboard. Show that each switchboard in the stack is capable of having its common-control subsystem control the stack of switchboards by processing a few line-to-line calls with each switchboard as master. Each changeover may be accomplished without powering down by switching the master board to the slave mode first. Another switchboard in the stack shall then be placed in the master mode.

4.6.5.3 Operator call extension in stack. In placing of some of the calls specified in 4.6.3.2, process them through the operator keyset of the master mode switchboard to terminations of other switchboards of the stack, to demonstrate interconnectivity and controllability of links between switchboards. Observe the usual tone, lamp, and connections of each call.

4.6.5.4 Automatic call extension in stack.

4.6.5.4.1 Dual-tone multifrequency receiver pool capacity. With three subsets of the DTMF line type from each switchboard in the stack, simultaneously place calls to three other subsets in other switchboards in the stack, and check for the usual tone, lamp, and connections of each call extension.

4.6.5.4.2 Link interconnectivity and capacity. The procedures specified in 4.6.1.2.2 and 4.6.1.2.3 shall suffice for demonstrating the ability of the stacked configuration to interconnect links and control their setups and breakdowns in automatic call extensions.

4.6.5.5 Trunk hunting without directory. Utilizing the configuration of terminations specified in Table XIII and the procedures specified in 4.6.2.4.2, establish calls to all six trunk hunting groups.

4.6.6 Operation under service conditions. The following tests shall be run before, during, and after (as appropriate) those tests specified in 4.6.10 and after the burn-in tests of 4.4.5 to demonstrate functionally a cross-section of terminal functions, call processes, and power/ringer functions.

(a) The 30-line switchboard shall be configured to include ten subsets and six trunks. The subsets shall include four 2-wire DTMF lines, four 2-wire rotary dial lines, and two 2-wire ringdown lines. The trunks shall include two 2-wire DTMF trunks, two 4-wire converter trunks, and two 6-wire E&M auto DTMF trunks. The trunk pairs can be wired in back-to-back configuration so as not to require external test equipments. Classmark in memory two DTMF lines and two rotary dial lines as precedence subscribers. The following tests shall be run:

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(b) Place five two-party calls from type II PCB terminations to the five other line subscribers, observing the normal lamp, service tone, ringing, and connection reactions of the switchboard in each call.

(c) Go off-hook with a priority precedence subscriber and place a priority call to an already busy terminal, observing all appropriate tone and lamp responses of the switchboard.

(d) Place two calls to the operator from one the the priority classmarked DTMF lines, one call as a 0 and one as an RO, to ascertain that requests for service are routed to the appropriate queue. (Do not call-answer.)

(e) Place a call from a DTMF line subscriber by dialing the 9 switch locator plan to a 4-wire converter trunk. Upon the second dial tone, dial a line termination. After answering the called line, observe that there is a voice connection.

(f) Repeat the above but with a 2-wire DTMF trunk. In all the above cases, observe at each step the proper lamp and service tone responses of the switchboard.

(g) Repeat the above but with a 6-wire auto DTMF trunk. In all the above cases, observe at each step the proper lamp and service tone response of the switchboard.

4.6.7 Compatibility. An equipment-integration test network shall be constructed, utilizing compatible equipments with which the switchboard is required to be operated. The compatible list of equipments to be used shall include those listed below, limited by the equipment loaned by the procuring activity:

(a) TA-312/PT Subsets

(b) TA-236/PT Subsets

(c) TA-838/PT Subsets

(d) SB-22 Switchboard, Manual

(e) SB-86 Switchboard, Manual

(f) CV-1919 Converter

(g) AN/TRC-166 Radio Terminal Set

(h) TH-85/GCC Telegraph-Telephone Terminal

(i) AN/TTC-38(V)1 or (V)2

(j) SB-3082(V)/GT Switchboard, Telephone Cordless Manual

(k) SB-22 with DTMF Applique (TA-955)

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(l) E&M Signal Converter (TBD)

(m) TA-312/PT with DTMF Applique (TA-977)

An integration test configuration such as that shown in Figure 1 shall be utilized to evaluate all operational integration features of the switchboard. Operational connections shall be made between each type of subset and each type of trunk equipment. Tandem operation of the switchboard shall be included. Prior to test initiation, the following setup shall be defined:

(a) Subscribers shall be classmarked with:

(1) Priority/routine precedences

(2) Intercept.

(b) Trunks shall be:

(1) Grouped according to switch locator

(2) Defined as normal or special.

Appropriate terminal-type PCB mixes shall be inserted. As a minimum, the following compatibility tests shall be employed to evaluate subscriber, operator, and system features:

(a) Routine calls involving operator action

(b) Priority calls involving operator action

(c) Conference calls to a distant switchboard and AN/TTC-38

(d) Routine call extensions to lines and trunks

(e) Priority call extensions to lines and trunks

(f) Preemption on lines and trunks

(g) Use of intercept facilities

(h) Recall utilization

(i) Use of trunk hunting

(j) Use of special and normal 20-Hz and converter, E&M trunks and DC closure trunks.

(k) Use of the switchboard as a tandem for converter/20-Hz, DC closure, E&M, and pabx trunk calls.

(l) Change of trunk groupings during switchboard use

(m) Use of insecure trunks.

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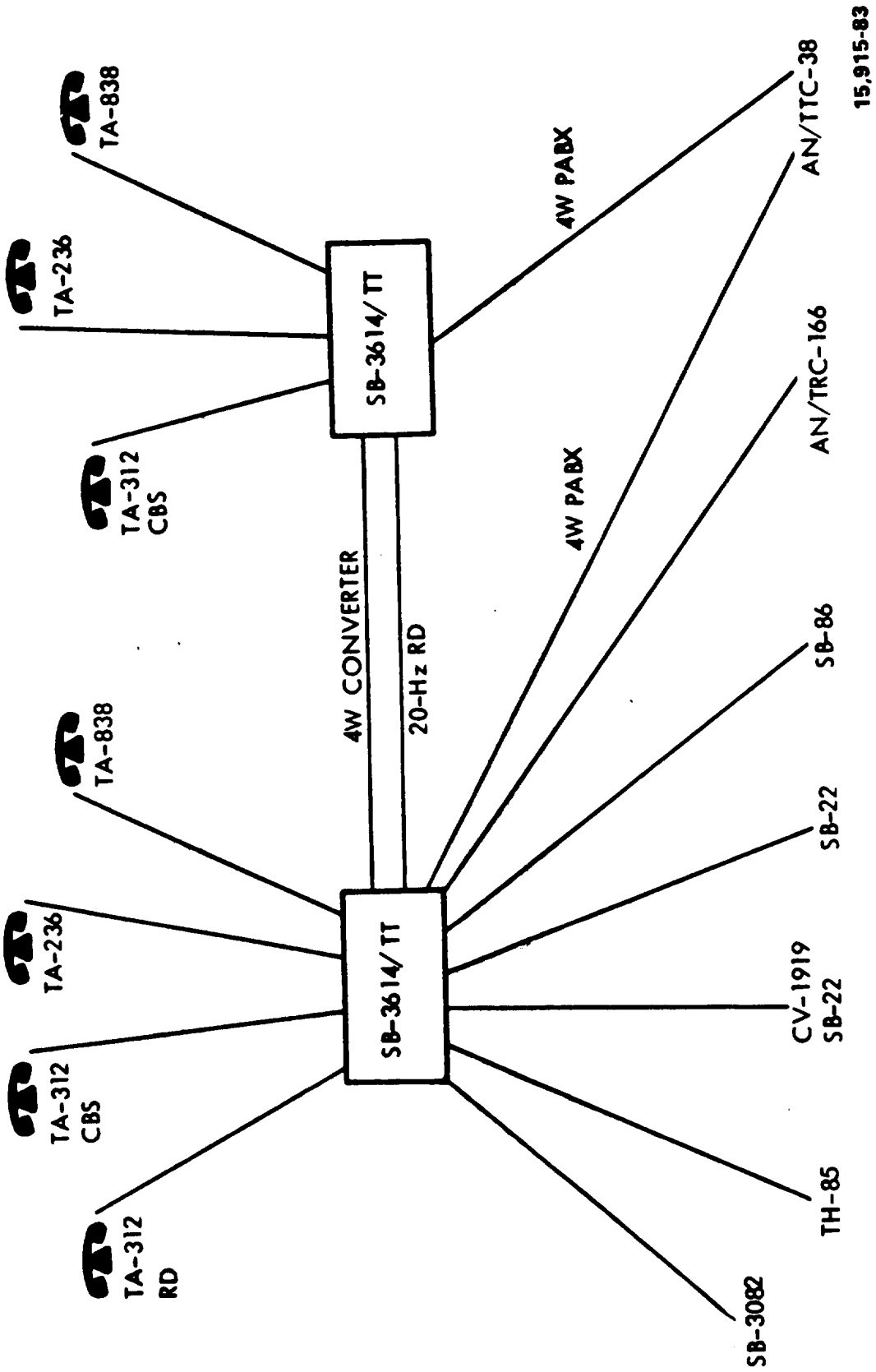


FIGURE 1. Typical compatibility network.

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4.6.8 Transmission. Transmission tests shall be performed as specified herein to verify switchboard compliance with 3.3.2.2 and 3.3.4. When terminal pairs involved in the tests are not terminated by test equipment, they shall be terminated in a 600-ohm ±1 percent resistor unless otherwise specified herein.

4.6.8.1 Terminal impedance. Establish a connection over a 2-wire path as shown in Figure 2. Connect up an impedance meter to terminal i while terminating terminal j in 600 ohms. Read magnitude and phase angle of the terminal impedance from the meter. Perform measurement at 300 Hz. Repeat the measurement at 500 Hz, 1,000 Hz, 2,000 Hz, 2,500 Hz, and 3,500 Hz. Repeat this measurement on each type of line or trunk termination.

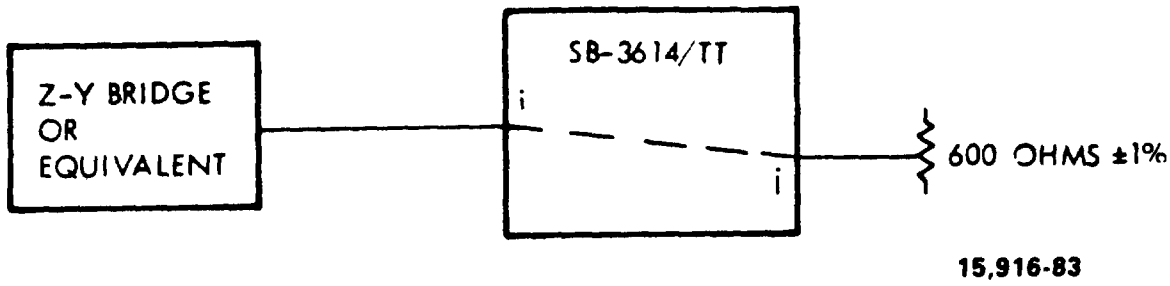


FIGURE 2. Terminal impedance.

4.6.8.2 Frequency response and insertion loss. Establish a connection in the switchboard between terminal i and j as shown in Figure 3. Connect a local oscillator to terminal i and a wave analyzer to terminal j. The wave analyzer in turn shall be connected to an XY plotter. With the signal level at terminal i set at -4 dBmW and the frequency initially set at 300 Hz, set the wave analyzer to sweep from 300 Hz to 108 kHz. Plot the response. Repeat this test using each different type and combination of line and trunk termination possible. When tests involve a 4-wire trunk termination, the frequencies swept shall be 0.3 kHz to 50 kHz.

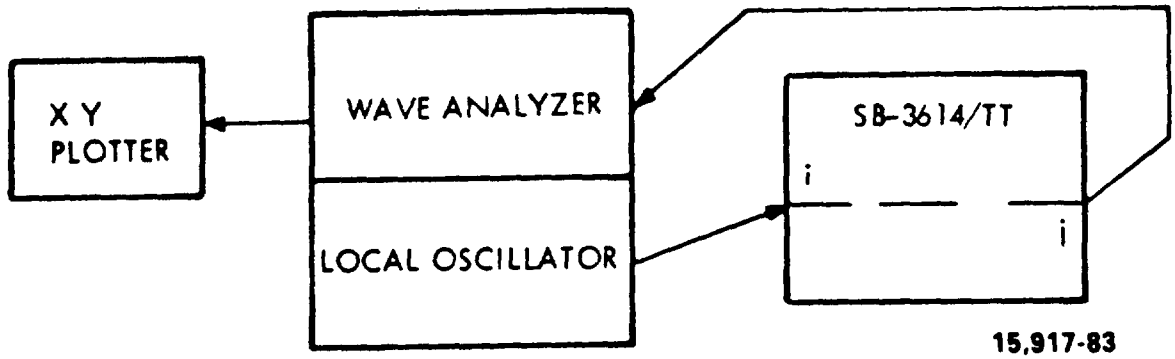


FIGURE 3. Frequency response insertion loss.

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4.6.8.3 Longitudinal balance. Establish a connection in the switchboard between terminals i and j as shown in Figure 4. Connect the local oscillator, vacuum tube voltmeter and test coil as shown. With V_1 at -4 dBmW and the frequency set at 300 Hz, record V_1 and V_2 (dB). The longitudinal balance is defined as follows:

$$dB_{REA} = dB_{(V1)} - dB_{(V2)}$$

Repeat the measurement at 500 Hz, 1,000 Hz, 2,000 Hz, 2,500 Hz, and 3,500 Hz. Repeat for each type of termination.

4.6.8.4 Harmonic distortion. Establish a connection in the switchboard between terminals i and j. Connect an audio oscillator to terminal i and a distortion analyzer to terminal j. With oscillator output adjusted to -4 dBmW at 300 Hz, observe the analyzer measurements. Repeat the measurement at 500 Hz, 1,000 Hz, 2,000 Hz, 2,500 Hz, and 3,500 Hz. Repeat for each type of line or trunk termination.

4.6.8.5 Envelope delay distortion. Establish a connection in the switchboard between terminals i and j as shown in Figure 5. Connect the test equipment to the terminals. Make delay measurements over the frequency range 300 Hz to 108 kHz. Measurements shall be performed utilizing each type of 2-wire termination.

4.6.8.6 Crosstalk. Establish connections in the switchboard between terminals i and j and k and l as shown in Figure 6. Connect the test equipment to the terminals as indicated. Terminals i and k shall be colocated on one termination PCB as shall be j and l. The matrix links used to connect through the switchboard shall be adjacent links. Perform the tests from 300 Hz to 3,500 Hz with terminations j and l terminated in 600 ohms. From 3,500 Hz to 108 kHz, change the terminations to 150 ohms. The local oscillator shall be set at a signal level of -4 dBmW as measured across terminal k. The measurements shall be repeated for each type of 2-wire line or trunk termination type.

4.6.8.7 20-Hz crosstalk. Connect a noise-measuring test set to termination i as shown in Figure 7. Terminate j with 5,000 ohms and connect the 20-Hz generator within the switchboard to the 5,000 ohms. The generator output shall be 135 V peak $\pm$5 percent at 20 Hz $\pm$5 percent. Measure the line output in termination i. Repeat measurement for each type of termination.

4.6.8.8 Noise level. Establish a connection through the switchboard between terminals i and j as shown in Figure 8. Connect up the test equipment and make the noise measurement. Repeat the measurement for each type of line or trunk termination.

4.6.8.9 Impulse noise. Establish a connection between terminals i and j in the switchboard and connect an impluse-counter measuring set to j. Set the threshold level of the measuring set to 50 dBrn. Successively make and break 10 separate connections. Record the counter reading at the completion of all calls.

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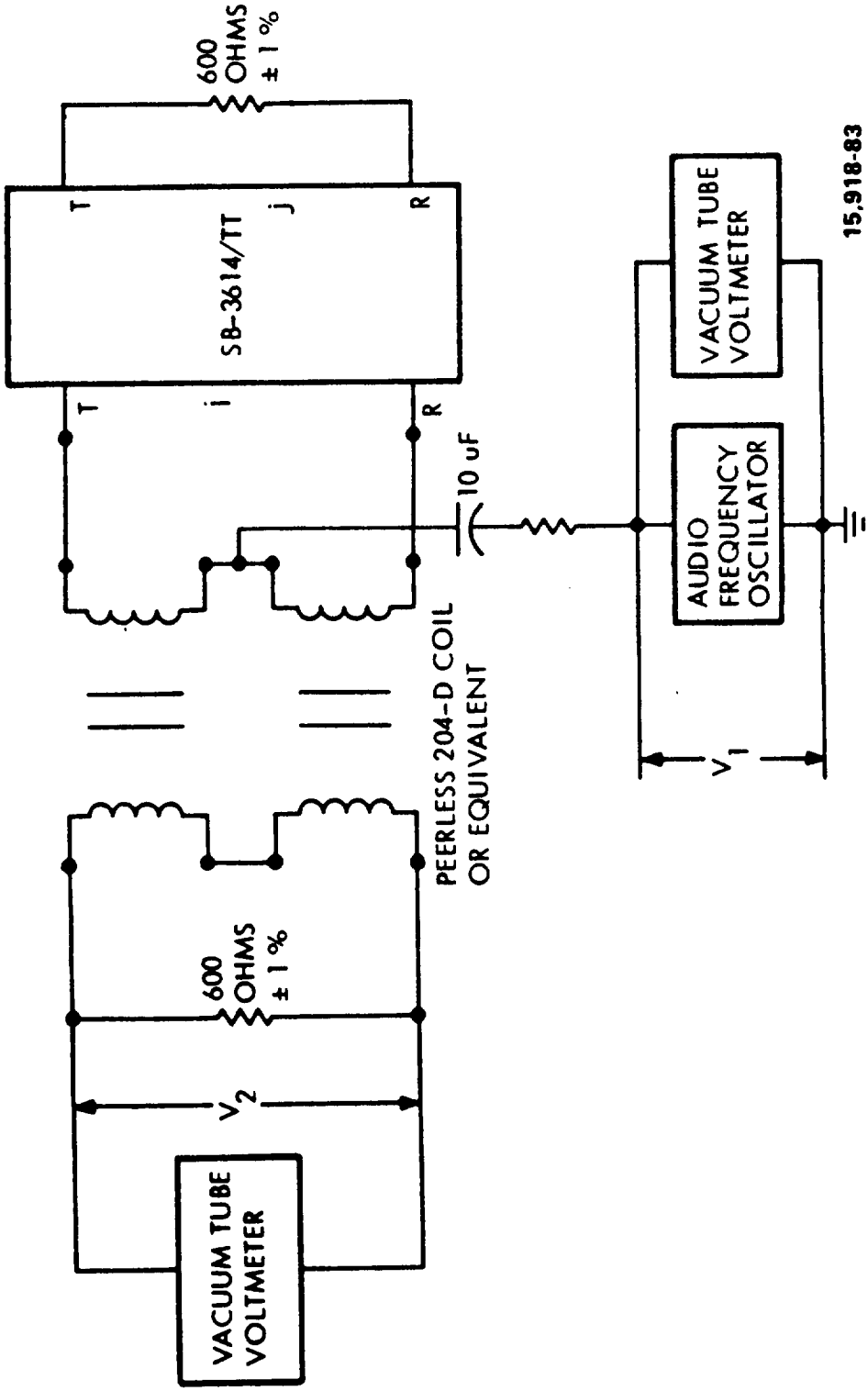


FIGURE 4. Longitudinal balance (REA method).

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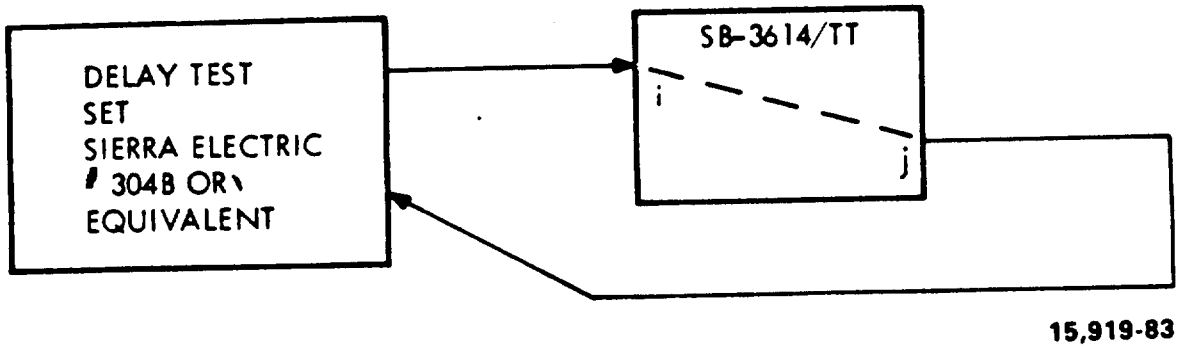


FIGURE 5. Envelope delay distortion.

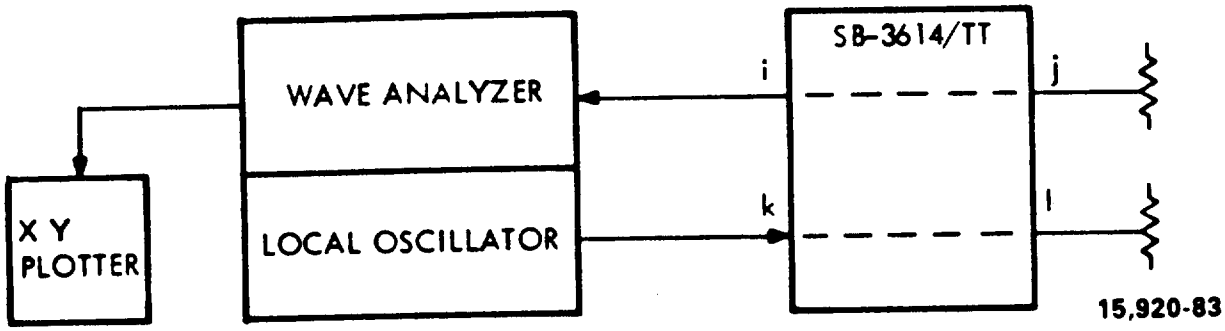


FIGURE 6. Frequency crosstalk.

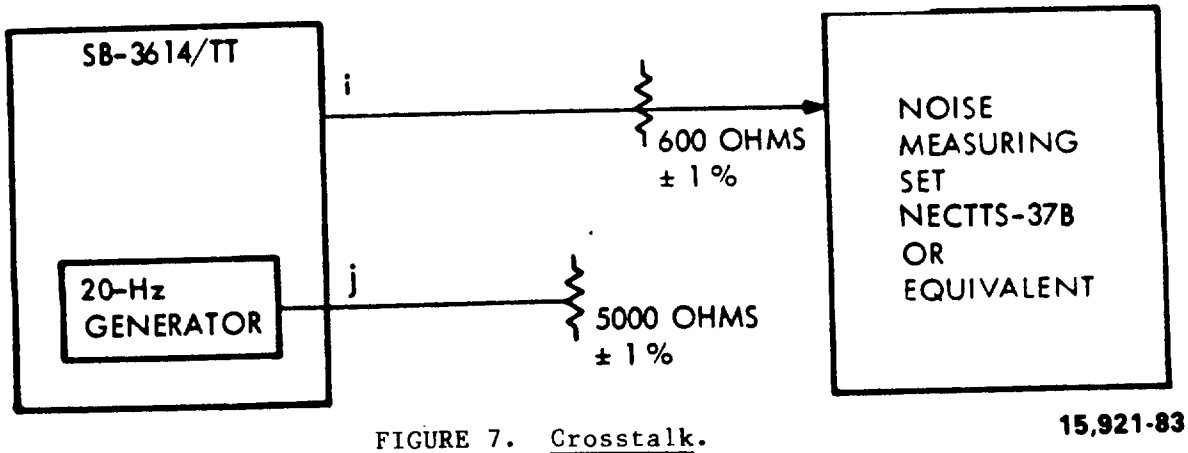


FIGURE 7. Crosstalk.

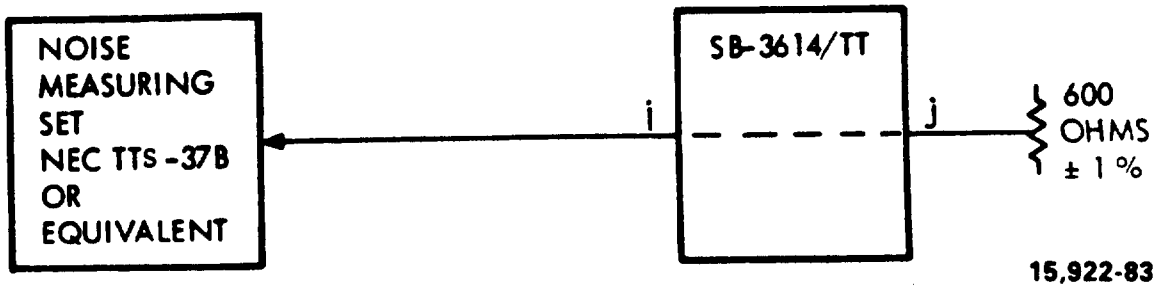


FIGURE 8. Noise level.

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4.6.8.10 Absolute delay. Establish a connection between i and j, connecting up test equipment as shown in Figure 9. Set the oscillator frequency at 300 Hz and the signal level as measured at i at -4 dBmW. Measure the delay between the signal at the oscillator terminal and at the switchboard rear panel connection point to terminal j. Repeat the test for frequencies of 500 Hz, 1,000 Hz, 2,000 Hz, 2,500 Hz, and 3,500 Hz. Repeat the test for 2-wire to 2-wire connections, 2-wire to 4-wire connections, and 4-wire to 2-wire connections.

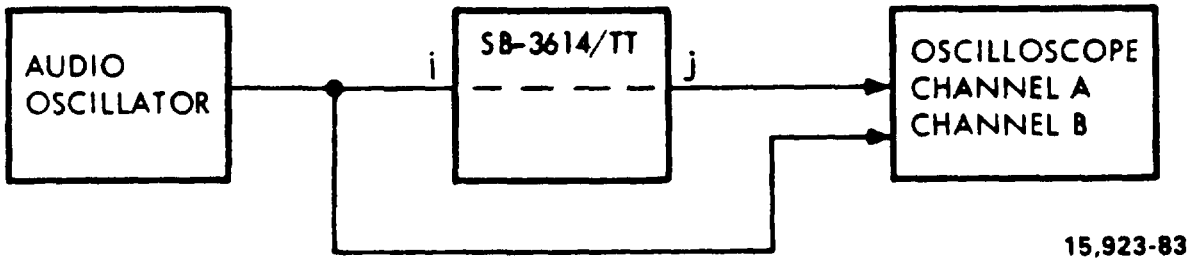


FIGURE 9. Absolute delay.

4.6.8.11 Conference circuit. Establish a connection between terminal i and j through the switchboard conference bridge. Connect an audio oscillator to terminal i and a signal-level measuring set to j. With the oscillator at 1,000 Hz and the signal level at -4 dBmW measured at terminal i, measure the signal level at terminal j. Repeat the measurement at j after establishing three more conferees on the bridge. Repeat the measurement a third time with five additional conferees on the bridge for a total of 10 connections. Each conferee connection shall be terminated in 600 ohms for the measurement.

4.6.8.12 Transhybrid loss. Establish a connection in the switchboard between a 2-wire termination, i, and a 4-wire termination, j. Connect up the test equipment as shown in Figure 10. With the oscillator set at -4 dBmW and frequency at 300 Hz, record the measurement at the 4-wire transmit pair (switchboard transmit) of terminal j. Repeat the test for frequencies 500 Hz, 1000 Hz, 2000 Hz, 2,500 Hz, and 3,500 Hz.

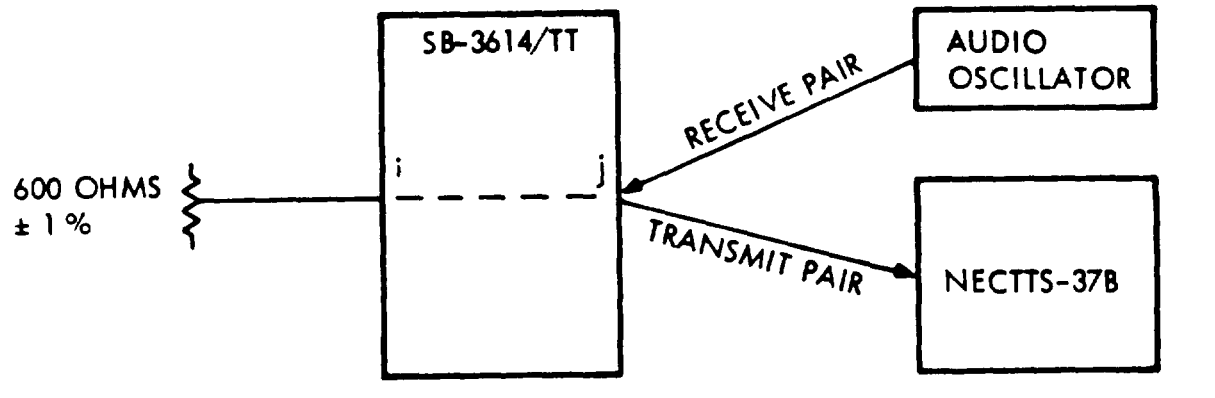


FIGURE 10. Transhybrid loss.

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4.6.8.13 Tone generators. Measurements shall be made on all common-equipment generators in the switchboard. Measurements shall be made so that frequency signal levels, modulation and interruption rates and their respective tolerances are determined. Generated tones to be measured are:

(a) all operator keysender tones

(b) seize (2250 Hz)

(c) sz ack/rl ack (570 Hz)

(d) restart (852/1209 Hz)

(e) release (2600 Hz)

(f) dial tone

(g) ringback

(h) wait-in-queue

(i) preempt

(j) error tone

(k) busy

(l) off-hook warning

(m) test tone.

(n) insecure warning tone.

Appropriate action shall be taken to assure that generated tones persist for sufficient time to allow accurate measurement. All measurements shall be made at the rear panel terminal binding posts, except that the test tone shall be measured at the test tone terminals.

4.6.8.14 20-Hz generator. Connect terminal j of the switchboard to a variable resistor load, a precision voltmeter, and frequency-meter test equipment. Initiate a ring cycle to the terminal j load, causing 20 Hz across the load. With the load set at greater than 50,000 ohms, measure the frequency and ring signal level. Repeat the test while reducing the variable load to 5,000 ohms, 1,000 ohms, 600 ohms, 100 ohms, and short circuit. At 100 ohms and short circuit, note that the 20-Hz failure lamp lights. After tests at 100 ohms and short circuit, repeat the test at 1,000 ohms load and note that frequency and signal levels are at their earlier level.

4.6.8.15 Detectors. Measurements shall be made on all detector types. Detectors shall be examined for sensitivity to frequency, signal level, and time.

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4.6.8.15.1 Tone detectors. Utilizing a common battery, DTMF type termination, go off-hook and receive dial tone. By appropriate techniques, prevent the DTMF detector from timing out. Replace the DTMF subset by an audio oscillator, precision voltmeter, and frequency meter as shown in Figure 11. Set the oscillator signal level at 0 dBmW and the frequency at 697 Hz -1.3 percent. Monitor the DTMF detector digital output and determine that detection is complete. Repeat the test at frequencies of 697 Hz and 697 Hz +1.3 percent, and with a signal level of -20 dBmW. Perform the same tests as above on the following tone detectors:

(a) All tone digit detectors on common equipment

(b) Interdigit detection on 4-wire trunk terminations

(c) Seize, release, and acknowledge detection on 4-wire trunk terminations.

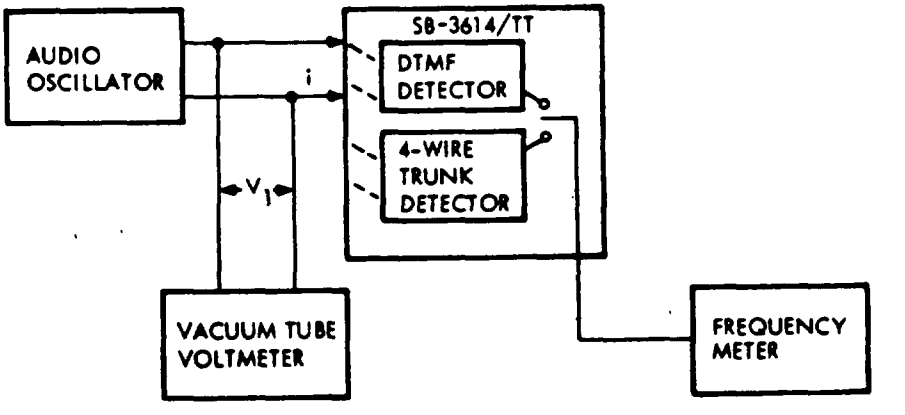


FIGURE 11. Tone detection sensitivity.

Perform the detection sensitivity tests on two DTMF receivers and three 4-wire trunk terminations. With a tone of 2,400 Hz and a signal level of 0 dBmW, no seize or release detection shall occur.

4.6.8.15.2 DC closure line detectors, CBS terminations. Connect a TA-312 CBS subset to terminal i of the switchboard through a resistive network as shown in Figure 12. Go off-hook and determine that terminal i's status lamp is flashing. Perform this test with prime voltage at 30 VDC. Repeat with voltage at 22 VDC. Repeat the above on terminals j, k and l, using different TA-312 subsets.

4.6.8.15.3 DC closure line detectors, CB termination. Connect a 450-ohm load to termination i. A 26 mA minimum current shall be measured with a prime voltage of 24 VDC. Replace the load with a variable resistor and ammeter in series. Slowly reduce the value of the resistance and note that the termination-status lamp i remains off as long as the ammeter is 10 mA or less. Reduce the resistance further and note that the status lamp turns on steadily when the current flow is greater than 10 mA and less then 16 mA. Perform this test with prime voltage at 24 VDC. Repeat the above tests with connection to terminals j, k, and l. Connect a Telephone Set, TA-938 to 1-3/8 miles or equivalent of WD-1 field wire to termination i. Recordings will be made as prime voltage is varied from 22 VDC to 30 VDC.

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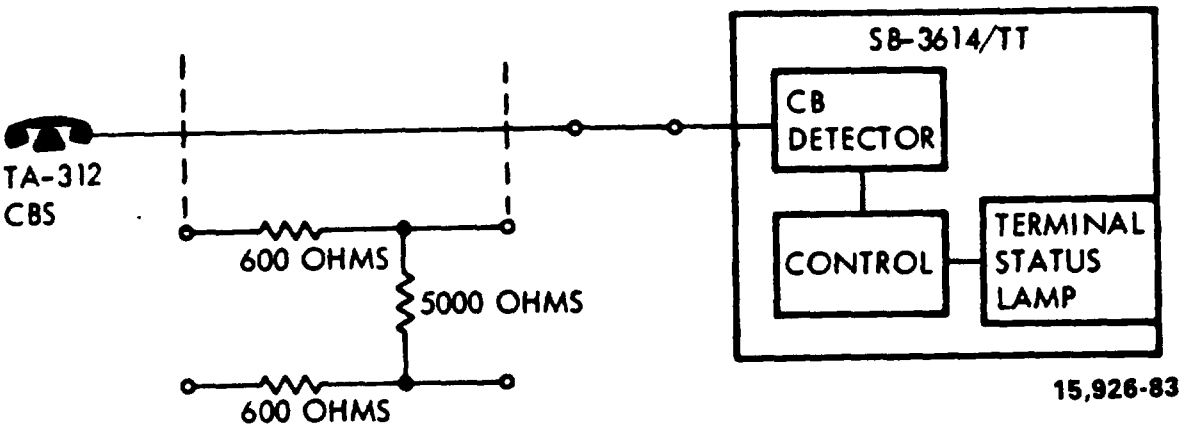


FIGURE 12. DC closure detection - CBS.

4.6.8.15.4 20-Hz detectors. Connect an oscillator to terminal i, a 20-Hz ringdown line termination circuit. Monitor the voltage and frequency impressed on the line. With the frequency set at 16 Hz and signal level at 12 V rms, note that the terminal status lamp i flashes slowly as in queue. Repeat the test with frequencies 20 Hz and 25 Hz on terminal i. Repeat the test on terminals j, k and l. Repeat the above test using a DC closure trunk termination circuit.

4.6.8.15.5 Release and seize detector timing. Using an oscillator, an attenuator, and an oscilloscope, measure the time required to detect release tone on a 4-wire tone-trunk PCB. This shall be determined by monitoring the transmit pair for acknowledge tone. Set the release signal at 2,600 Hz -1.3 percent and -20 dBmW for this test. Repeat at 2,600 Hz +1.3 percent. Determine the length of time for acknowledge to stop after release has ended. These tests shall also be performed where seize 2,250 Hz is employed instead of release.

4.6.8.15.6 Acknowledge detector timing. Utilizing an oscilloscope, an oscillator, and frequency and voltmeter test gear as in Figure 13, initiate a release cycle from a 4-wire trunk termination i of the switchbboard. Setting the acknowledge tone as provided in the oscillator at 570 Hz -1.3 percent and -20 dBmW, determine the detection time of the 570 Hz by monitoring for the time it takes the release to stop after acknowledge has started. If necessary, use appropriate means to prevent release time out. Repeat the test with the frequency set at 570 Hz +1.3 percent. Test should also be performed on terminals j, k, and l.

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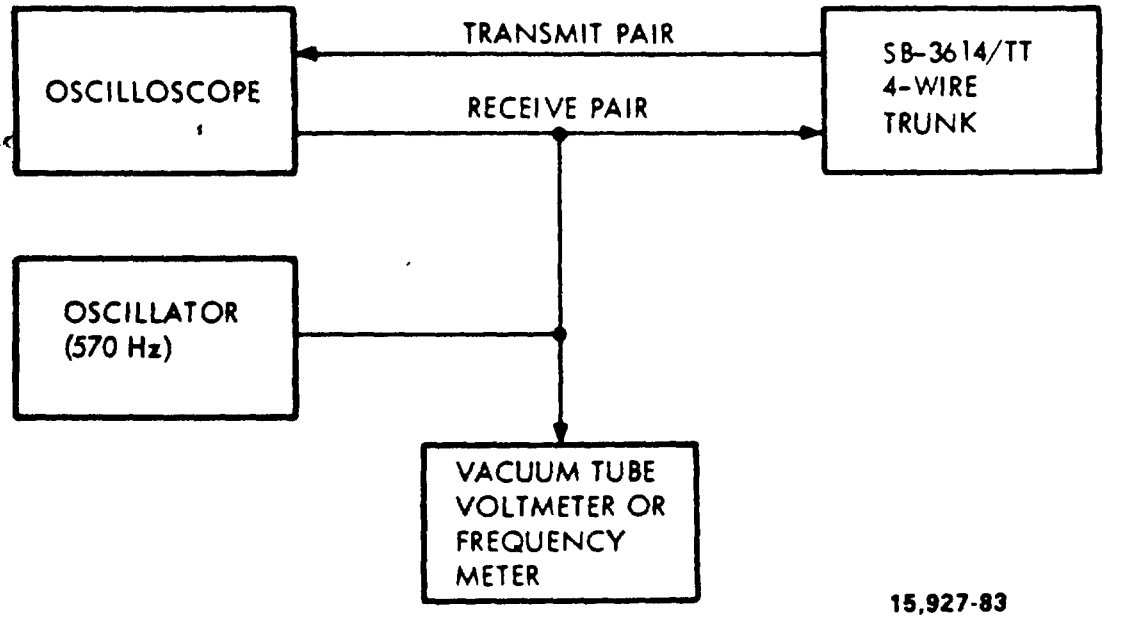


FIGURE 13. Acknowledge detection time.

4.6.8.15.7 E&M lead detectors. Connect a decade box between ground and the E lead of terminal i, a E&M trunk termination circuit. Monitor the status of the termination while adjusting the decade box. Notice that an OFF-HOOK condition will exist when resistance to ground is less that 1000 ohms and that an ON-HOOK condition exists when the resistance to ground is greater than 20,000 ohms. Repeat the test for terminals j, k, l.

4.6.8.16 Rise time test. Establish a connection over a 2-wire path with terminal i, a 2-wire terminal and terminal j, a 4-wire terminal. With a square wave generator set at 16 kHz and connected to terminal i, measure with an oscilloscope the rise time of the signal at the transmit terminal j. Repeat the procedure with the square wave generator connected to the receive terminal j and the oscilloscope connected to terminal i. Terminations during measurements are 600 ohms.

4.6.8.17 Return loss test. On any 4-wire termination on both receive and transmit pairs, measure the return loss using the following method on an idle tereminal. With a transformer bridge technique, establish a reference level across the bridge from a sine wave generator set at 300 Hz. All four sides of the bridge shall have 600-ohm terminations. The reference level shall be set to -4 dBmW as read across cne 600-ohm resistor. Substitute the path to be measured for that resistor and measure the level returned across the opposite side of the bridge. Repeat this procedure for 500 Hz, 1 kHz, 3.5 kHz, 10 kHz, 30 kHz and 50 kHz.

4.6.8.18 Maximum signal test. Establish a connection 2-wire to 4-wire. Using a square wave generator set at 16 kHz, and connected to the 2-wire terminal and with the 4-wire transmit terminal terminated in 600 ohms, observe both input and output signals on an oscilloscope. Adjust the signal level until a change on the input does not change

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the output. Record the input level. Reverse the position of the 2-wire and the 4-wire terminals and repeat the test.

4.6.9 Mechanical tests.

4.6.9.1 Weight. The switchboard shall be weighed in its transport configuration including headset, termination box, and power and interconnect cables, excluding ground rod and cable. In this configuration, the switchboard weight shall not exceed 66 pounds.

4.6.9.2 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.5.1. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect rating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners or separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking-pin fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.6.9.3 Visual and mechanical inspection. Visual and mechanical inspection shall be in accordance with MIL-STD-252, and shall be applied to the following: parts, mechanical construction, loose nuts and screws, application of specified treatments and miscellaneous defects. The equipment and controls shall be inspected for proper mechanical and electrical operation. Wiring, soldered connections, grounds, welds, finishes, etc. shall be inspected for workmanship. Clearances, dimensions and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance to approval facsimilies where facsimilies are required. Examination may also be made for other similar visual and mechanical defects which are contrary to the requirements specified herein.

4.6.10 Environmental tests.

4.6.10.1 High temperature. The switchboard shall be subjected to high temperature test in accordance with Method 501, Procedure II, of MIL-STD-810. For step 4, the chamber temperature shall be 130°F for test-item operation. Before, during, and after exposure to high

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temperature, the tests specified in 4.6.6 shall be performed on the switchboard.

4.6.10.2 Low temperature. The switchboard shall be subjected to low temperature test in accordance with Method 502, Procedure 1, of MIL-STD-810. The storage temperature (Step 2) shall be -65°F, and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (Step 4) shall be -25°F. Before, during, and after exposure to low temperature, the tests specified in 4.6.6 shall be performed on the switchboard.

4.6.10.3 Immersion. The switchboard and transit case, as prepared for field transport, shall be subjected to immersion test in accordance with Method 512, Procedure 1, of MIL-STD-810. At completion of this test, the switchboard and transit case shall show no evidence of leakage.

4.6.10.3.1 Air seal test. The switchboard and transit case shall be opened and closed again in such a manner as to break and remake the seal. Immediately thereafter, the switchboard, as field transported, shall be subjected to a vacuum of 2 pounds per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the case. The vacuum shall then be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period, the decrease in vacuum shall not exceed 0.01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily.

4.6.10.4 Rain. The switchboard in operating configuration shall be subjected to rain test in accordance with Method 506, Procedure 1 of MIL-STD-810. During each 30-minute rain period, wind shall be maintained for 15 minutes, commencing 5 minutes after the start of the period as specified in the procedure. The tests specified in 4.6.6 shall be performed during the last 10 minutes of each 30-minute period.

4.6.10.5 Salt fog. The switchboard and transit case shall be subjected to salt fog test in accordance with Method 509, Procedure 1, of MIL-STD-810. Operation of the switchboard shall not be required, and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the equipment surfaces which can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

4.6.10.6 Humidity. The switchboard and transit case shall be subjected to humidity test in accordance with Method 507, Procedure III of MIL-STD-810. In Step 7, the chassis shall be removed from its enclosure or the test item shall be otherwise opened so as to expose the normally sealed areas to the chamber environment. Before exposure to humidity, during the last 5 hours of Step 6, and at the conclusion of the test, perform the tests specified in 4.6.6.

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4.6.10.7 Altitude. Perform the test specified in 4.6.6, and then subject the switchboard and transit case to altitude test in accordance with Method 500, Procedure I, of MIL-STD-810. This test shall be performed with the switchboard in its normal condition for air shipment, and no operation of the switchboard shall be required. Following this test, the switchboard shall be placed in its operating condition and shall be returned to the conditions of Step 3 of Procedure I. At this time, the tests of 4.6.6 shall be repeated on the switchboard.

4.6.10.8 Fungus. The switchboard and transit case shall be subjected to fungus test in accordance with Method 508, Procedure I, of MIL-STD-810.

4.6.10.9 Vibration and shock.

4.6.10.9.1 Vibration. The switchboard and transit case in transit condition shall be subjected to test in accordance with Method 514.1, Procedure XI, Part 2, of MIL-STD-810. Before and after this test, operational tests specified in 4.6.6 shall be performed.

4.6.10.9.2 Shock. The switchboard in its transport configuration including headset, terminal box, and SB/S power cables, but not ground rod/cables, shall be subjected to shock tests in accordance with Method 516.1, Procedure II and V, of MIL-STD-810. Before and after each test, the operational tests specified in 4.6.6 shall be performed.

4.6.10.10 Sand and dust. The switchboard in operating configuration and transit case shall be subjected to sand and dust test in accordance with Method 510, Procedure 1, of MIL-STD-810. Step 3 shall be performed immediately after reaching stabilization in Step 2. Before exposure to dust and during Step 5, perform the tests specified in 4.6.6.

4.6.10.11 Sunshine. The switchboard shall be exposed to the sunshine test in accordance with MIL-STD-810, Method 505, Procedure II. Before and after test, the operational tests specified in 4.6.6 shall be performed.

4.6.10.12 Static load. The switchboard and transit case shall be subjected to a static load test. Two hundred (200) pounds shall be evenly distributed over a 4-inch by 12-inch area of the top and bottom of the cases. Covers shall be on. Before and after static load test, the operational tests specified in 4.6.6 shall be performed on the switchboard. Inspect the switchboard and transit case for any mechanical damage or deformation.

4.6.11 Reliability tests.

4.6.11.1 Reliability qualification test. The first articles shall be tested in accordance with 4.3. The reliability testing shall be conducted in the sequence specified in MIL-STD-781. The test shall be in accordance with the requirements of 3.4.

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4.6.11.1.1 Reliability room temperature. This test shall be conducted at 70°F ±3.6 F. A total of 2,470 randomly distributed calls shall be placed through the switchboard, with a maximum of 14 lost calls permitted. If 15 or more calls are lost or 5 consecutive lost calls occur, the test shall cease and positive corrective action, consisting of design change, modification, or rework actually related to observed failures, shall be applied to the switchboard (corrective action applies to the other first articles when they possess the same deficiency as the one which failed). No rerun shall be made on the switchboard until the corrective action is completed. All reruns shall begin from call number one. Failure of the switchboard to pass the reliability test after three attempts shall mean that the switchboard has not met the requirements of 3.4.1, and this shall be considered cause for rejection of the switchboard.

4.6.11.1.2 Reliability low temperature. The same requirements specified in 4.6.11.1.1 shall apply except for the following: the test shall be conducted at 25°F ±3.6°F. A total of 3,280 calls shall be placed with a maximum of 14 lost calls permitted. If 15 or more lost calls or 5 consecutive lost calls occur, the test shall cease and positive corrective action shall be taken. As specified in 4.6.11.1.1, three attempts shall be allowed before the switchboard is rejected.

4.6.11.1.3 Reliability high temperature. The same requirements specified in 4.6.11.1.1 shall apply, except for the following: the test shall be conducted at 130°F ±3.6°F. A total of 1,640 calls shall be placed with a maximum of 14 lost calls permitted. If 15 or more lost calls or 5 consecutive lost calls occur, the test shall cease and positive corrective action shall be taken. As specified in 4.6.11.1.1, three reruns shall be allowed before the switchboard is rejected.

4.6.11.2 Production reliability sampling tests. Switchboards for this test shall be tested at high and low temperatures. When, during the performance of any part of this test, the quantity of lost calls equals or exceeds the reject criteria, the test shall cease and no rerun shall be made on the same or a new sample from the same lot, unless positive action, consisting of design change, modification, or rework actually related to observed failures, has been applied to the lot. All reruns shall begin from call number one. Failure of the switchboard to pass any part of the reliability sampling test after three attempts shall mean that the switchboard has not met the requirements of 3.4.1, and this shall be considered cause for rejection of the switchboard.

4.6.11.2.1 Production reliability high temperature test. The production high-temperature reliability test shall be conducted in accordance with Figure 14 and Table XVI, at 130°F ±3.6°F. A series of random calls shall be made, with none of the equipment removed or busied out prior to commencing the tests, and no repairs or busy-out shall be allowed during the test. The grade of service provided by common equipment shall not be reduced for the duration of the test. The test shall be conducted until the number of lost calls versus

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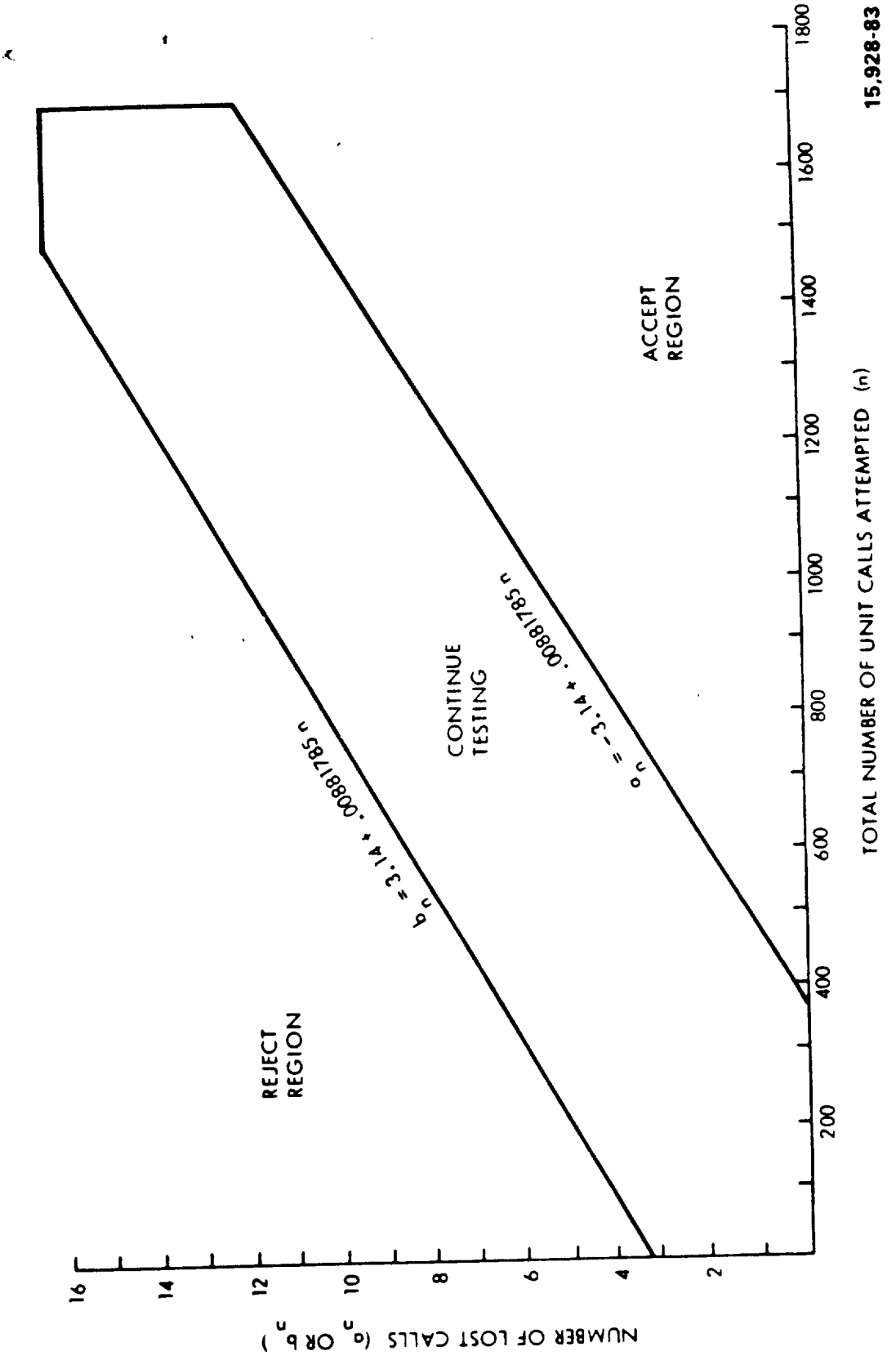


FIGURE 14. High temperature call test; ambient temperature range $130^{\circ}F \pm 3.6^{\circ}F$.

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TABLE XVI. High-temperature call test.

Number of lost calls	Reject	Continue test	Accept
0	Not applicable	0-359	360
1	Not applicable	0-469	470
2	Not applicable	0-584	585
3	Not applicable	0-699	700
4	4-100	101-809	810
5	5-215	216-924	925
6	6-325	326-1039	1040
7	7-440	441-1149	1150
8	8-552	553-1264	1265
9	9-665	666-1377	1378
10	10-780	731-1431	1432
11	11-832	833-1604	1605
12	12-1006	1007-1699	1700
13	13-1120	1121-1699	1700
14	14-1234	1235-1699	1700
15	15-1346	1346-1699	1700
16	16-1700	Not applicable	Not applicable

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attempted calls is such that it causes the results to move into the accept-or-reject region. No less than 600 unit calls shall be performed per 8-hour test period. When, during the performance of any part of this test, the quantity of lost calls causes the results to move into the reject region of Figure 14, the test shall cease and positive corrective action shall be applied to the switchboard. No rerun shall be made on the switchboard until the corrective action is completed. Subsequent reruns shall begin from the call number designated by the procuring activity representative. Failure of the switchboard to pass the reliability test after three attempts shall mean the switchboard has not met the requirements of 3.4.1, and this shall be considered cause for rejection of the equipment. Proper instrumentation shall be provided to insure immediate recognition of a lost call. Lost calls shall be as defined in 4.6.11.3.1. The types of calls shall be as described in 4.6.11.3.

4.6.11.2.2 Production reliability low temperature test. The low temperature production test (see 4.6.11.1) shall be conducted in accordance with Figure 15 and Table XVII, at 25°F ±3.6°F. The same requirements as in 4.6.11.2.1 shall apply, except for temperature range and number of lost calls as noted.

4.6.11.3 Fixed-length test. The fixed-length test shall be conducted with no equipment busied out prior to commencing the test. Neither preventive maintenance servicing nor repairs to failed equipment shall be made during this test. Shown in Table XVIII is a list of five different types of calls which, along with their respective frequencies of use, shall be required to make up the total call count of the fixed-length test. Any type of call listed in Table XVIII, even if by its nature it involves multiple connection such as conference, shall be counted as a single call. Tandem trunk calls which are simulated in "n" back-to-back loops shall be counted as "n+1" calls. A call shall be successful only when it is completed fully. For example, a conference call shall be designated a lost call if any of the connections are not completed; only when all connections are completed shall the call be considered successful. The power shall be shut off at least 2 hours per 24-hour period of testing. The rate of calls initiated shall be based on the following distribution:

(a) Calls equalling 20 times the termination capacity of the switchboard placed over a 4-hour operating period; this busy period shall only be conducted once per test.

(b) A minimum number of calls equal to the termination capacity of the switchboard per call hour during the rest of the test.

(c) All types of calls listed herein shall be placed during the 4-hour high-traffic period.

4.6.11.3.1 Lost calls. A call may involve any combination of circuits within the switchboard which are meant to work together. For example, any termination shall be able to work with any register, link, or common equipment to which it is meant to have access. Calls may be placed simultaneously or separately. A call shall be

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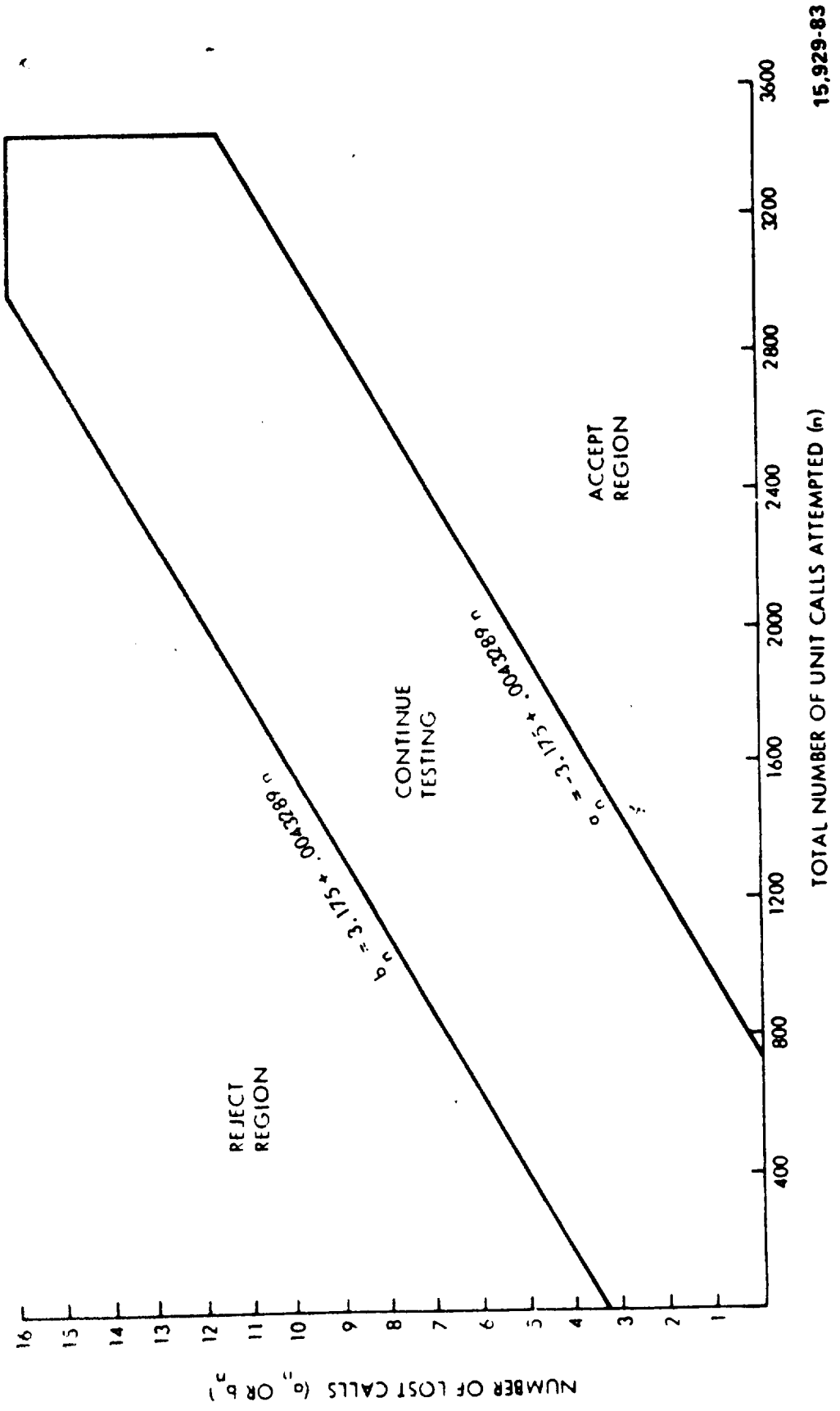


FIGURE 15. Low-temperature call test ambient temperature range: 25°F ±3.6°F.

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TABLE XVII. Low temperature call test.

Number of calls lost	Reject	Continue test	Accept
0	Not applicable	0-734	735
1	Not applicable	0-964	965
2	Not applicable	0-1199	1200
3	Not applicable	0-1429	1430
4	4-190	191-1659	1660
5	5-425	426-1889	1890
6	6-655	656-2119	2120
7	7-885	886-2354	2355
8	8-1120	1121-2584	2585
9	9-1350	1351-2814	2515
10	10-1580	1581-3044	3045
11	11-1810	1811-3279	3280
12	12-2040	2041-3449	3450
13	13-2270	2271-3449	3450
14	14-2505	2506-3449	3450
15	15-2735	2736-3449	3450
16	16-3450	Not applicable	Not applicable

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TABLE XVIII. Fixed-length reliability test call make-up.

Call types	Relative frequency of use
Local	60
Trunk (single link)	35
Conference	1
Preemptive	2
Recall	1

considered lost if any of the circuits in the switchboard do not perform as designed during the call, from the call origination to the call termination. Examples of lost calls are: degradation of audio, distortion, failure to send ring tone or ring-back tone, failure to release, excess crosstalk or interferences, failure to obtain dial tone, premature release, false busy signals. The transmission medium between the terminal equipment and the switching equipment may include any characteristics which the equipment is designed to handle. The terminal equipment may include any equipment with which the switchboard is meant to interface.

4.6.12 Electromagnetic compatibility tests. The switchboard shall be tested for compliance with the requirements of 3.9. Test setups and test procedures shall conform to general measurement techniques of MIL-STD-462.

4.6.12.1 Intrasystem compatibility. The switchboard shall be exercised through all its modes of operation as specified herein. The switchboard shall meet the requirements of MIL-STD-461, Notice 1, as provided for in Table XIX.

4.6.12.2 Bonds and grounds. Prior to performance of electromagnetic interference/electromagnetic compatibility testing, compliance with the requirements of 3.9.1 shall be ascertained.

4.6.13 Maintainability demonstration. In order to demonstrate compliance to the requirements of 3.6, the contractor shall perform a maintainability demonstration test in accordance with method 1 Test B of MIL-STD-471. The sample size shall be 30. The mean time to repair shall be 15 minutes. The producer's risk shall be 10 percent. Task selection shall be in accordance with Appendix A of MIL-STD-471. Accept/reject criteria shall be in accordance with the decision procedure specified in test method 1 of Appendix B of MIL-STD-471. Retest provisions shall be as follows.

4.6.13.1 Retest stage. A provisional schedule for special or repeat tests to investigate deficiencies or trouble areas. Deficiencies shall be corrected in any item which has failed to meet the acceptance criteria. The corrected portions of the item and any

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TABLE XIX. Emission and susceptibility tests.

REQUIREMENTS	FREQUENCY RANGE
CE01 1/	10 kHz to 50 kHz
CE02	10 kHz to 20 kHz
CE04	20 kHz to 50 MHz
RE02 2/ 3/	14 kHz to 1000 MHz
RS03 3/ 4/	14 kHz to 400 MHz

1/ Test method CE01 shall be performed over the frequency range of 10 kHz to 50 kHz.

2/ The upper frequency limit for this test shall be 1000 MHz.

3/ Each of 50 percent of switchboard terminals shall be connected to 100 feet of WD-1/TT, Telephone Cable, terminated in a 600-ohm load.

4/ The test shall be performed over the frequency range of 14 kHz to 400 MHz.

other portions of the item affected by the correction shall be retested during this stage. The maintenance tasks to be demonstrated shall be designated by the procuring activity.

5. PACKAGING

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be level A or C as specified in the contract or order in accordance with MIL-E-17555 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. This switchboard is intended to provide semi- and fully-automatic switching between 30 2-wire lines and trunks and between 2-wire local and 4-wire trunked automatic central office switches. When two or three switchboards are electrically connected, switching shall be provided for up to 60 or 90 terminations. Provisions have been included to facilitate interchange of pluggable termination PCB's, enabling the use of different types of field or fixed facility subsets and trunks required by service-to-service

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peculiar needs. The switchboard shall be deployable in a tactical environment and as a two-man carry item.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification

(b) Quantity of first articles required (see 3.1)

(c) Mix of termination PCB's required (see 3.3.2.1.1)

(d) When rough-handling test is required (see 4.5.1)

(e) Levels of preservation, packaging and packing (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporate a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.	Options
(a) 4.3	Produce Assurance Test Demonstration Evaluation Plan	DI-R-1701	
(b) 4.6.11	Burn-In Test Report	DI-R-7040	
(c) 4.4.3.1	Quality Inspection Defect Report Group C Failure	DI-R-1721	

(Data item descriptions related to this specification and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. First article testing shall be in accordance with Section 4, and shall be witnessed by a representative of the procuring activity.

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6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - MC
Project No. 5805-N316

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